# Introduction to Pangenomics

Presented by Vikram and Katie



# Outline

1. What is a pangenome?
2. Why do we need/want a pangenome?
3. Genome assemblies (the building blocks of pangenomes)
4. Representations of pangenomes
5. Examples of pan-genomes
6. Open questions in the field
7. What we will do with pangenomes in this course

# What is a pangenome?

- “The pan-genome represents the entire set of genes within a species, consisting of a core genome — containing sequences shared between all individuals of the species — and the ‘dispensable’ genome.” - https://www.nature.com/articles/d42859-020-00115-3

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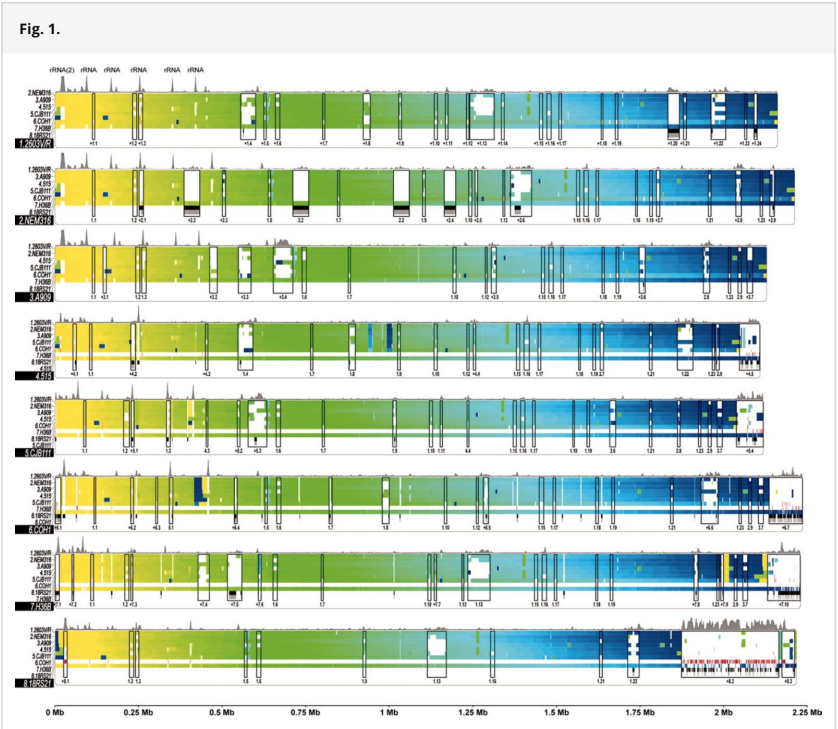
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- This definition is practical. We currently cannot, nor do we necessarily want to, capture the variation of every single individual in a species.
- Captures the goal of pangenomics, to understand common and most rare variation.
- Would change “species” to “clade”.
- Essentially, a pan-genome is a tool to facilitate comparative genomics.

# What is a pangenome?




A pangenome is a set of genes within a species

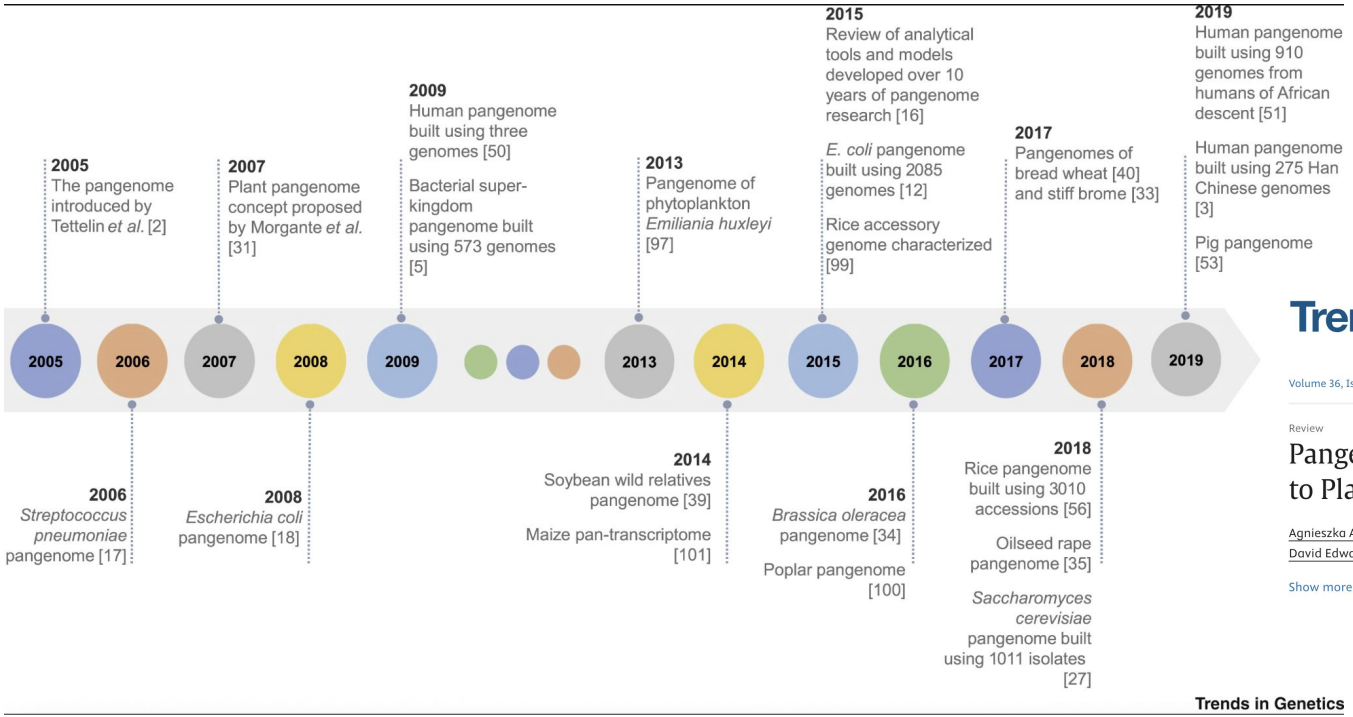
RESEARCH ARTICLE | GENETICS |

**Genome analysis of multiple pathogenic isolates of *Streptococcus agalactiae*: Implications for the microbial "pan-genome"**

Hervé Tettelin, Vega Masignani, Michael J. Cieslewicz, Claudio Donati, Duccio Medini, Naomi L. Ward, Samuel V. Angiuoli, Jonathan Crabtree, Amanda L. Jones, A. Scott Durkin, Robert T. DeBoy, Tanja M. Davidsen, Marirosa Mora, Maria Scarselli,

https://www.pnas.org/doi/10.1073/pnas.0506758102

# What is a pangenome?




**Trends in Genetics**

CellPress

Volume 36, Issue 2, February 2020, Pages 132-145

Review

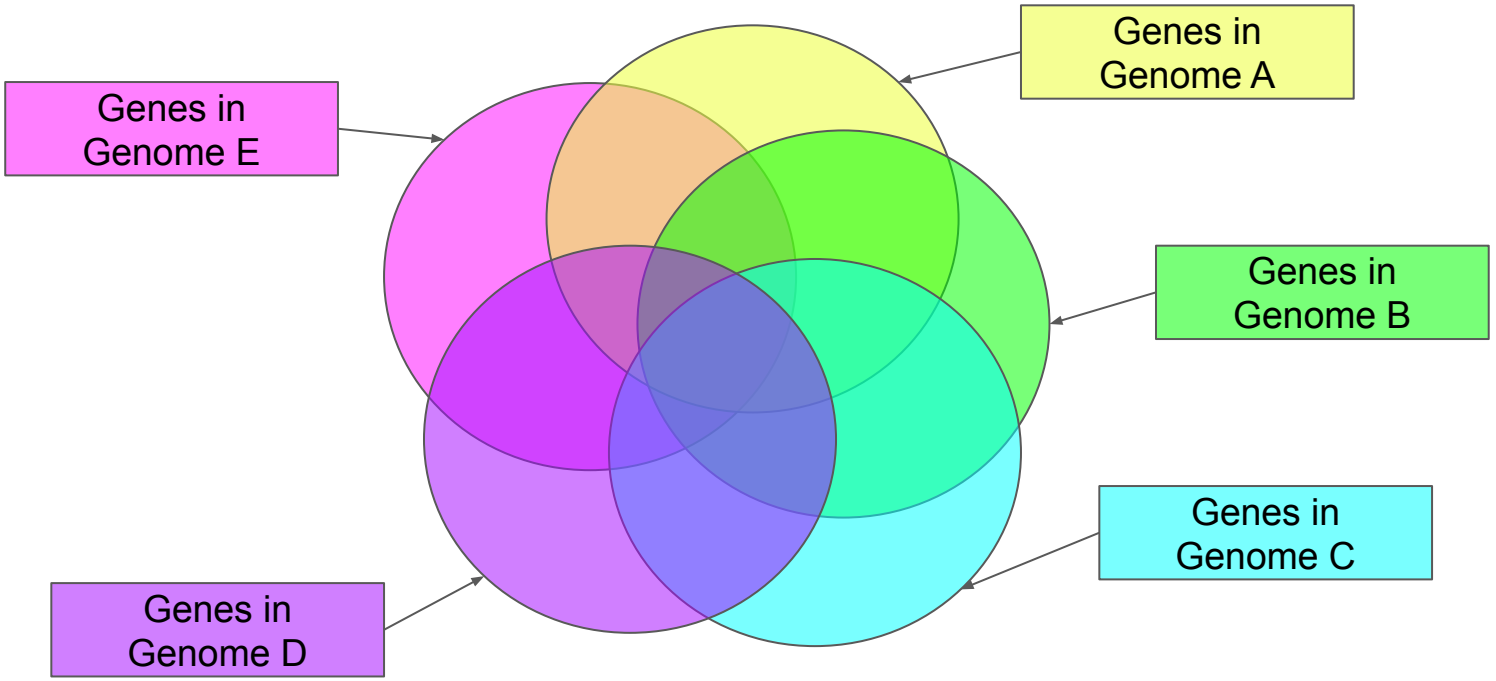
## Pangenomics Comes of Age: From Bacteria to Plant and Animal Applications

Agnieszka A. Golicz [1], Philipp E. Bayer [2], Prem L. Bhalla [1], Jacqueline Batley [2], David Edwards [2]

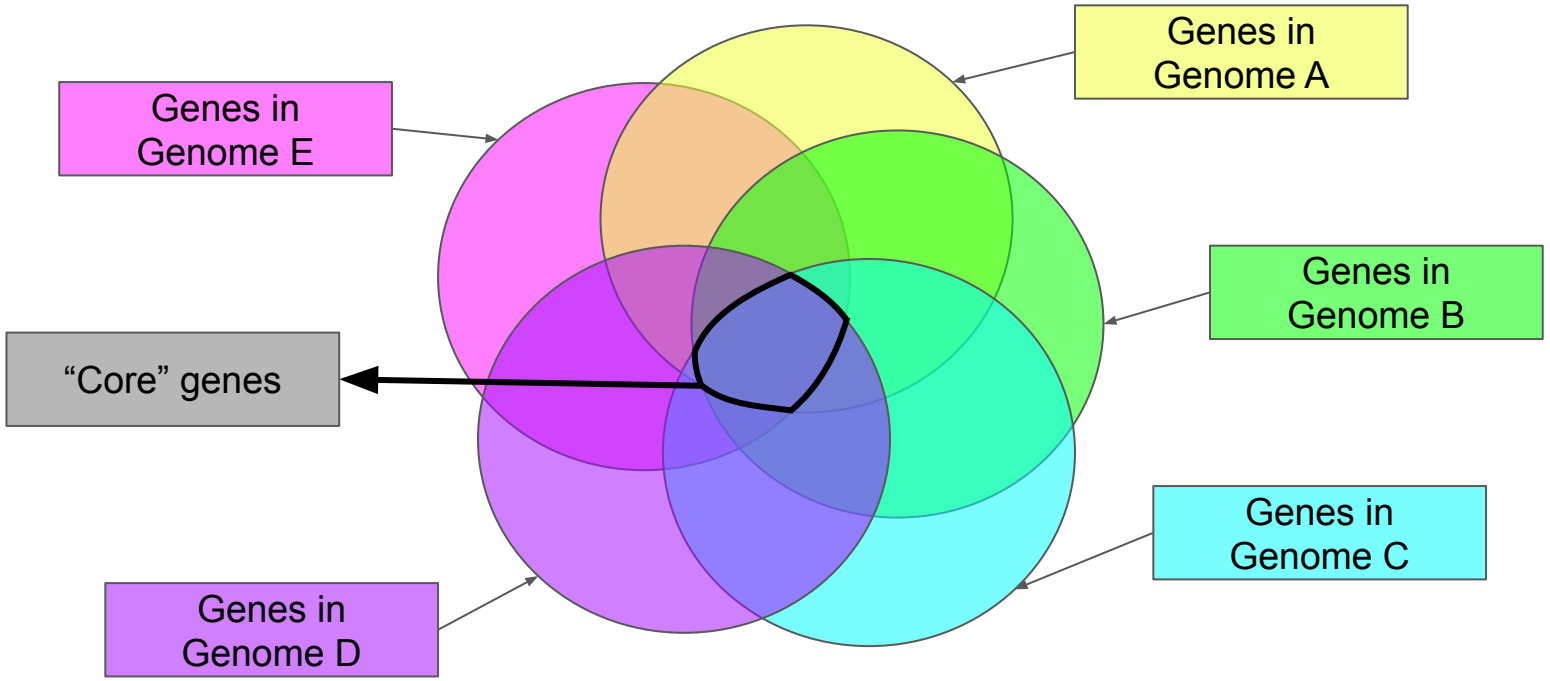
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https://www.sciencedirect.com/science/article/pii/S016895251930246X

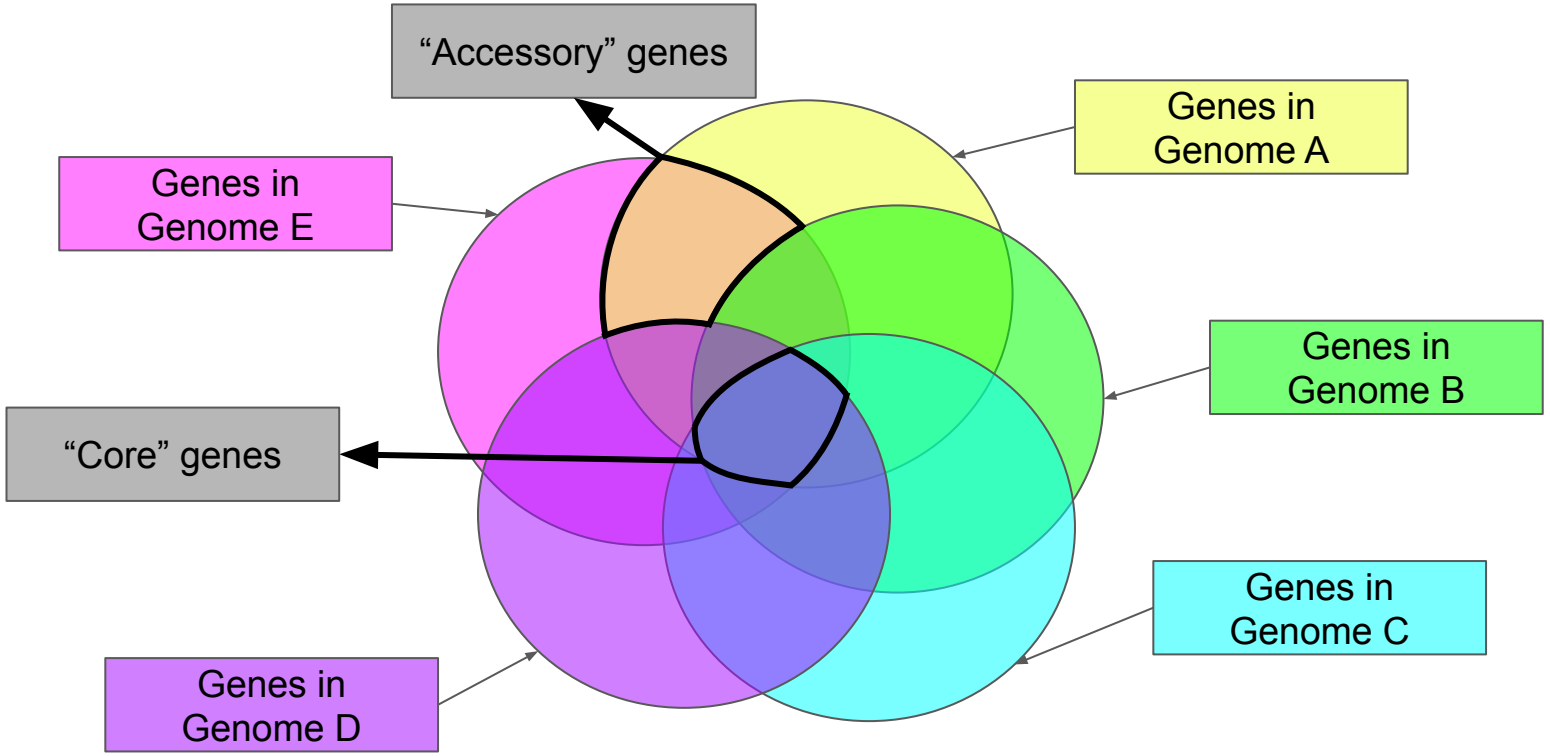
# Core vs. accessory vs. unique genes/sequences



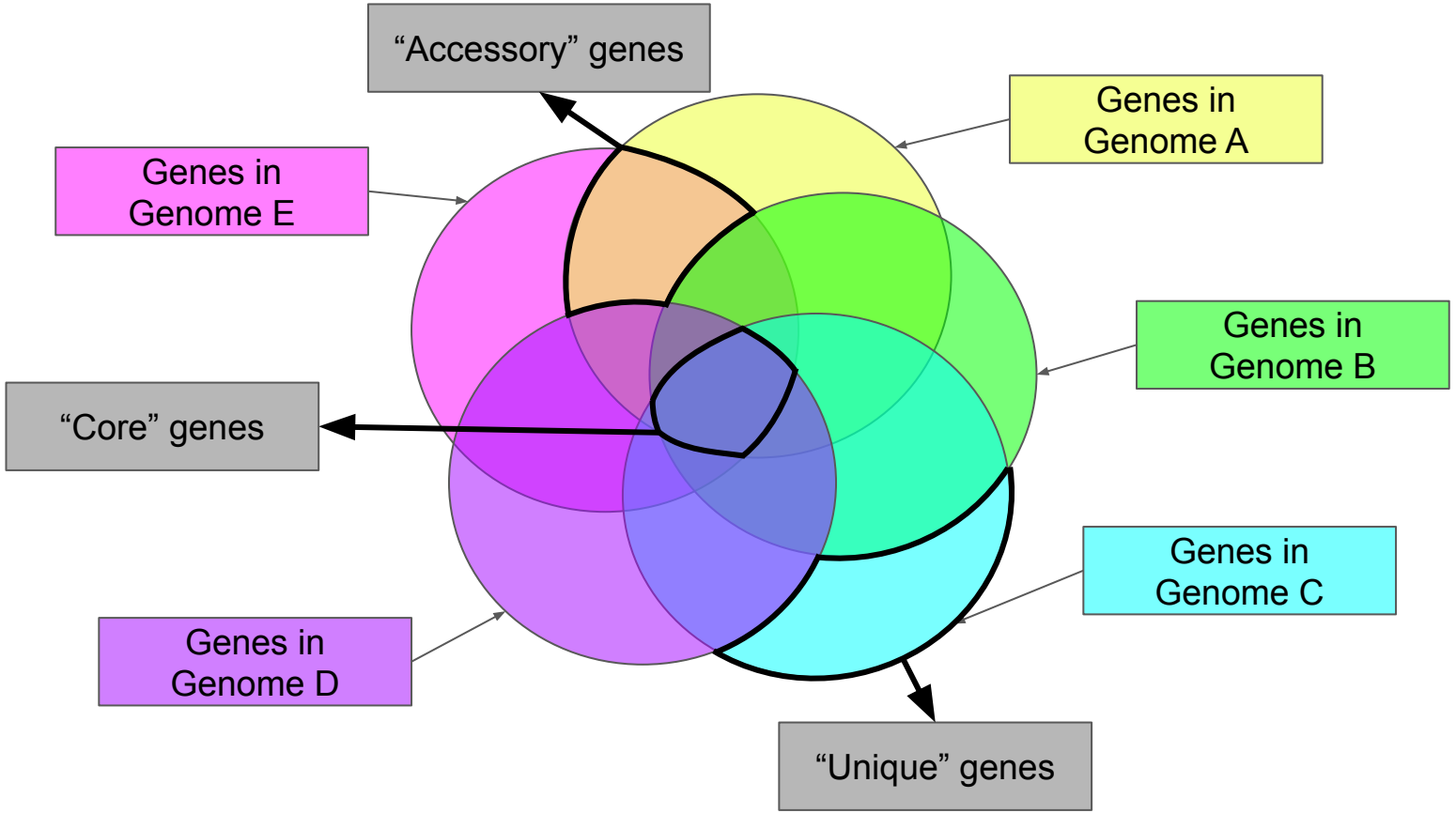
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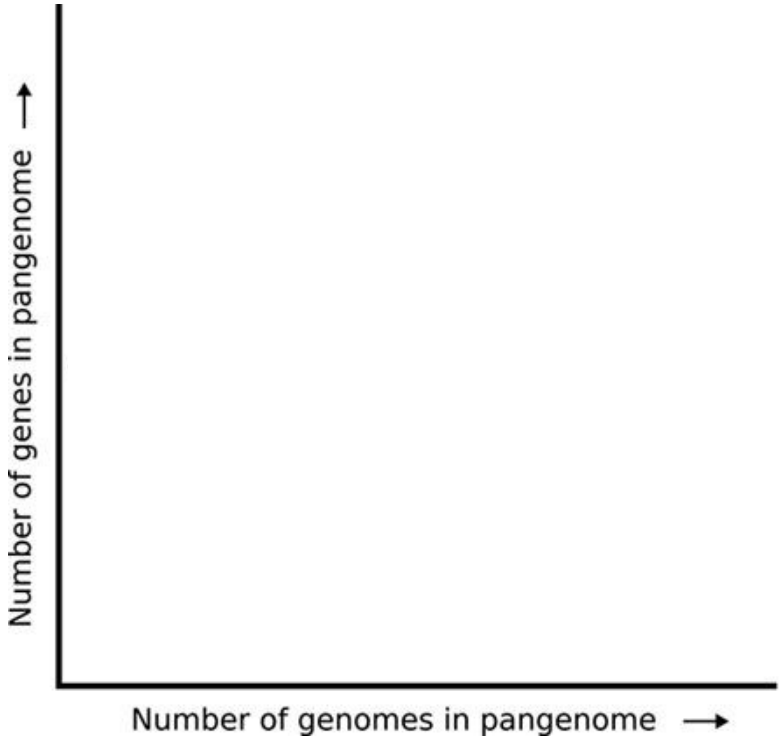
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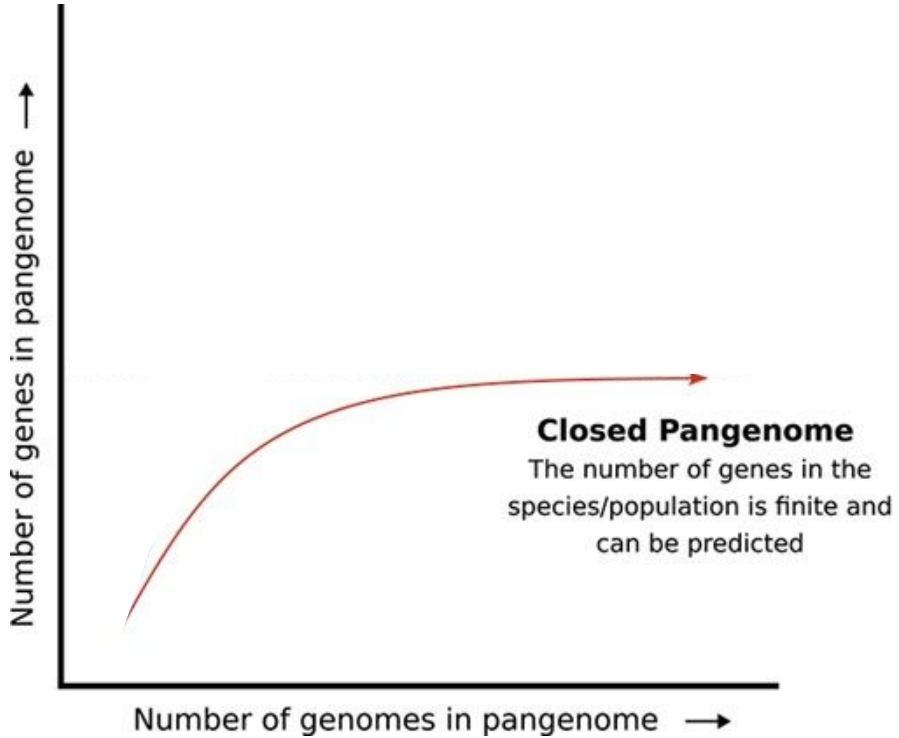


# Open versus closed pangenomes



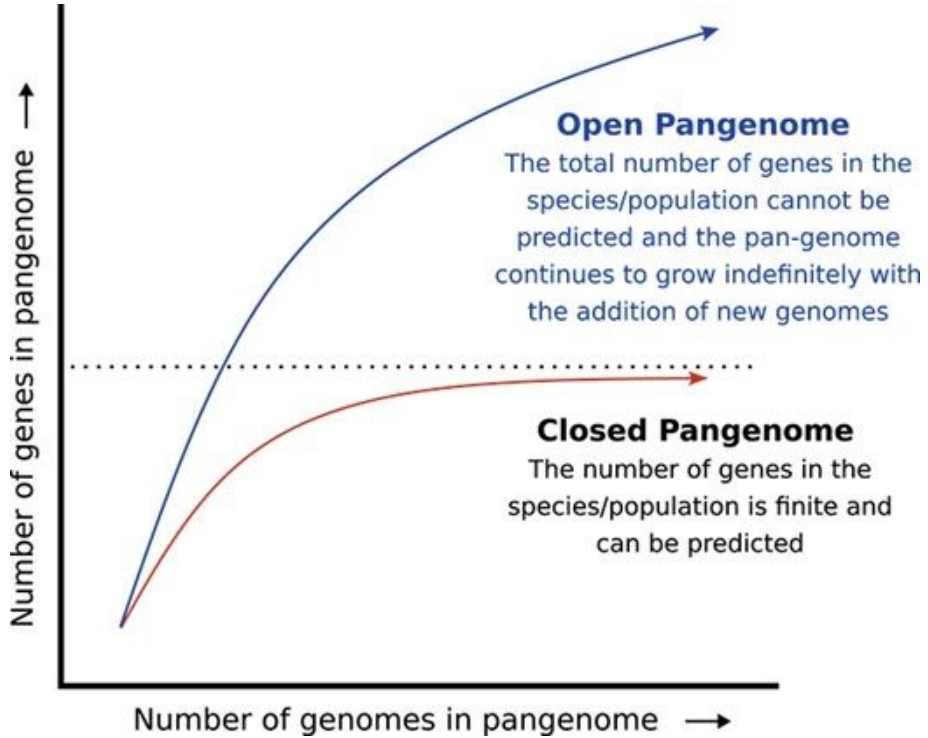

Matthews et al. 2024 https://academic.oup.com/bib/article/25/6/bbae588/7902219

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# What is a pangenome?

nature › technology features › article

TECHNOLOGY FEATURE | 18 April 2023

## Every base everywhere all at once: pangenomics comes of age

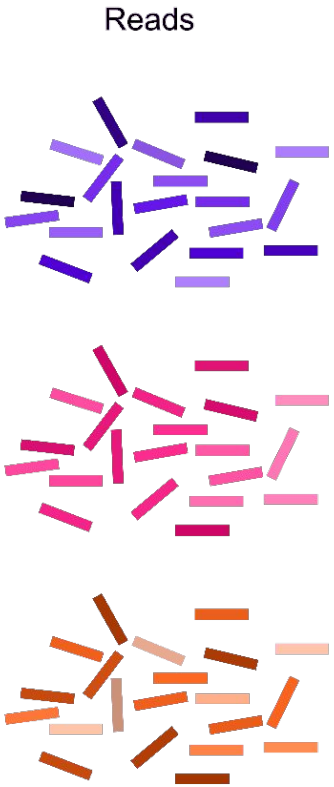
**Multi-genome assemblies called pangenomes can capture genetic diversity in a species, but researchers are still working out how best to build and explore them.**

By Michael Eisenstein

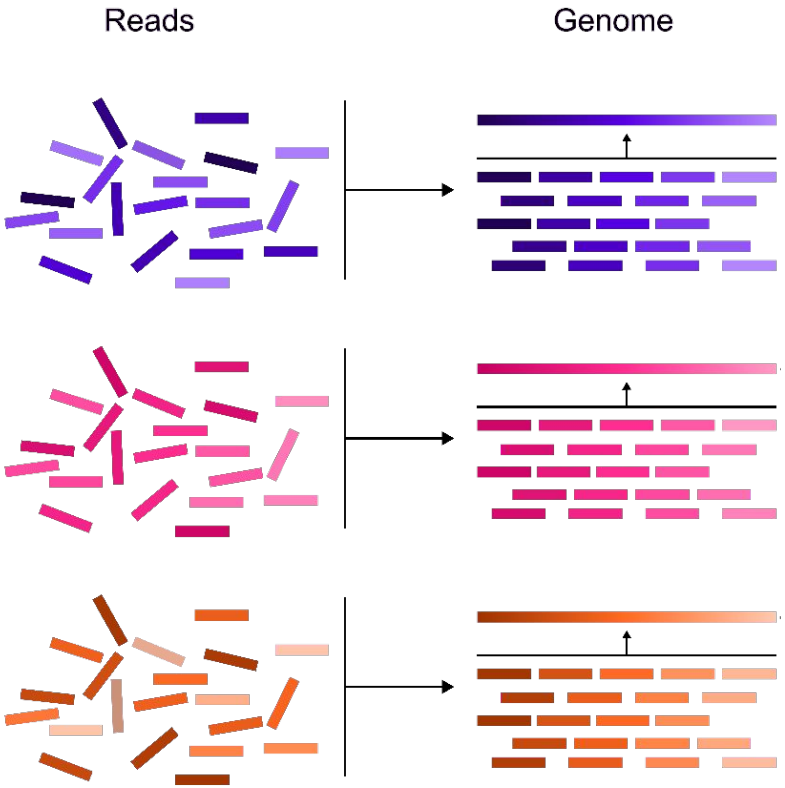



https://www.nature.com/articles/d41586-023-01300-w

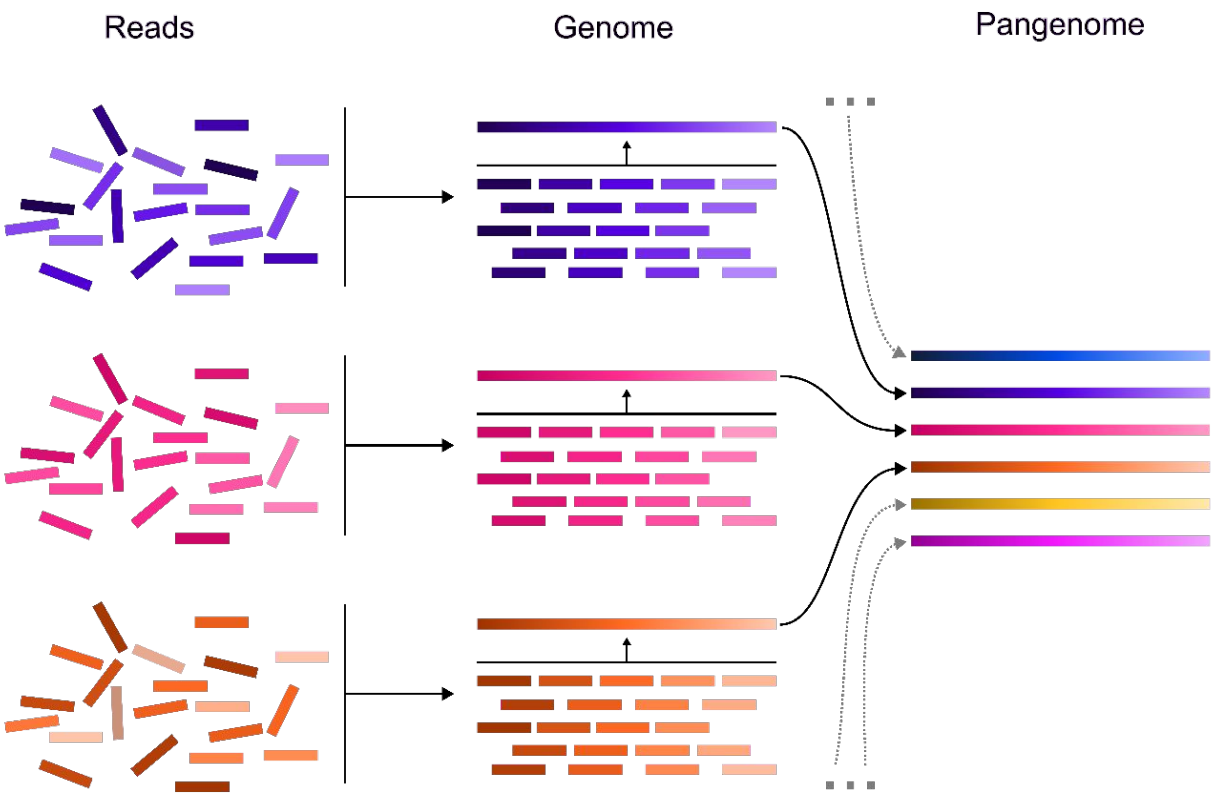
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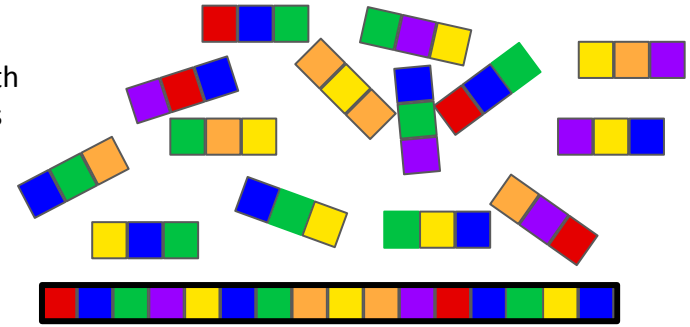


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Sequencing reads with a similar haplotypes

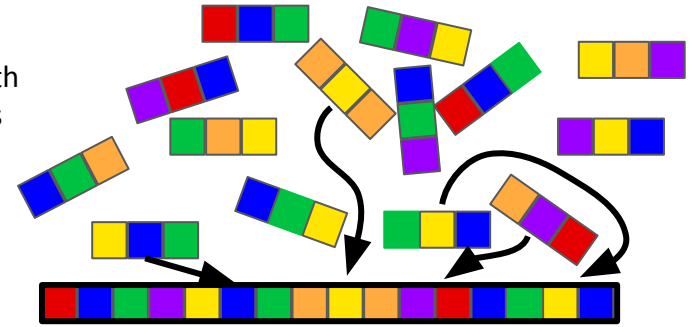


**Reference**

**One genome only represents one individual (at best)**

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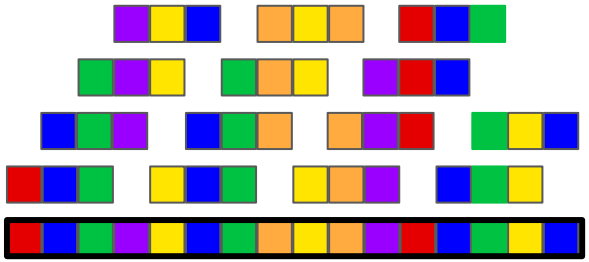


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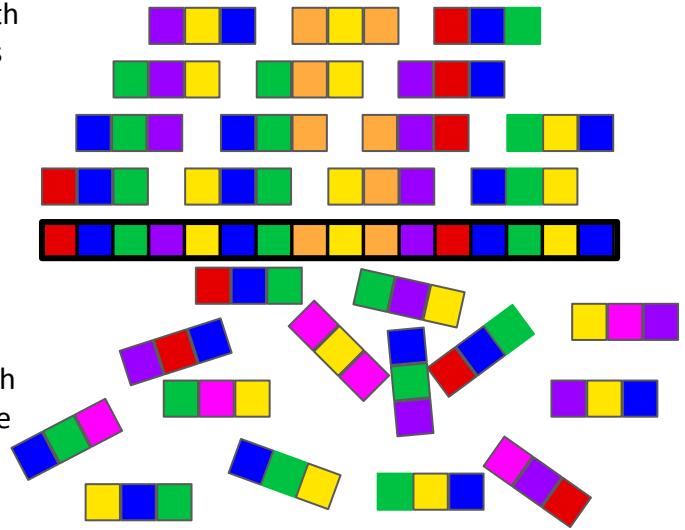
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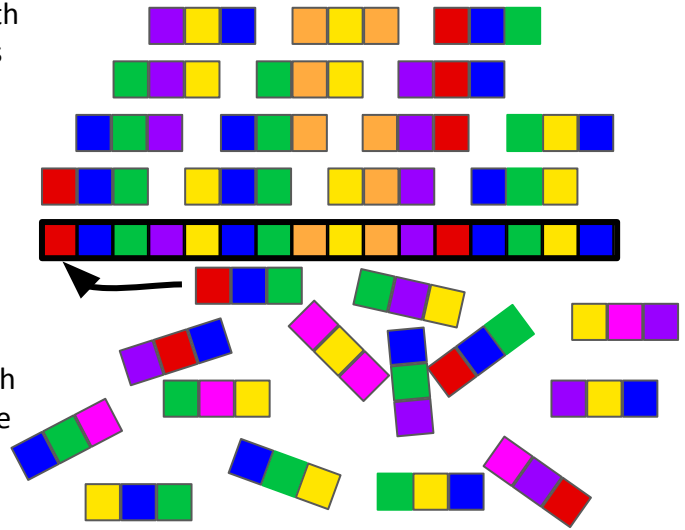
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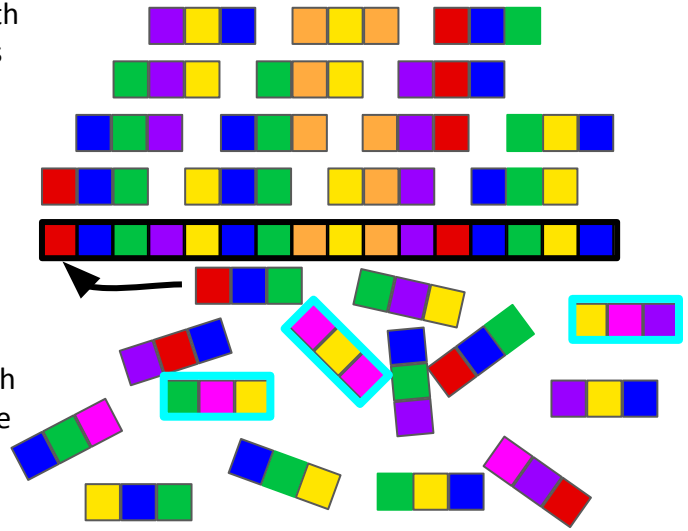
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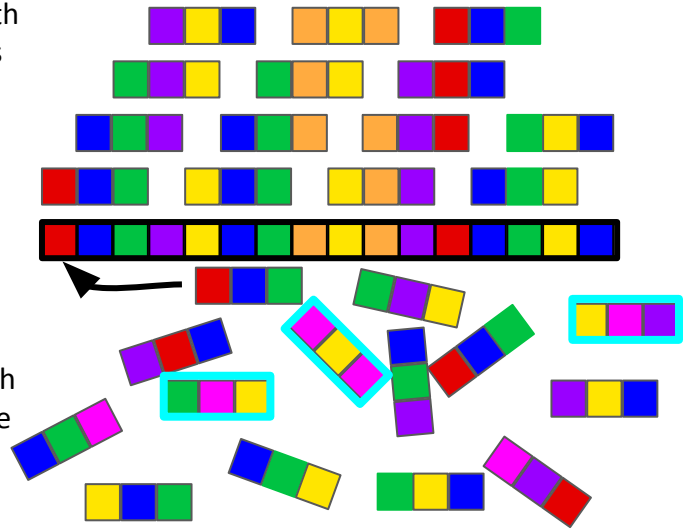
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*Canis lupus*



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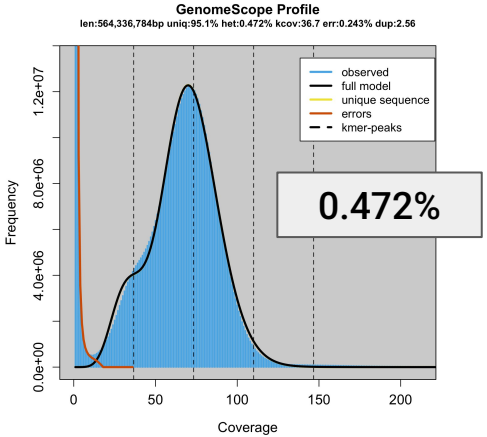
| Platform | Het (%) | Kcov (het. peak) | Error rate (%) | Model fit | Genome size est. | Bps with coverage < 2x |
|---|---|---|---|---|---|---|
| Element Aviti UltraQ | 0.25% | 34.0 | 0.06% | 0.82 | 2,908,208,748 | 2,180,013 |
| Illumina HiSeq 2x250, | 0.27% | 30.7 | 0.26% | 1.06 | 2,903,028,543 | 1,012,551 |
| PacBio Onso | 0.29% | 22.5 | 0.03% | 0.74 | 2,904,805,881 | 1 |
| PacBio Revio with SPRQ | 0.28% | 50.6 | 0.08% | 1.08 | 2,866,227,392 | |
| PacBio Revio HiFi | 0.23% | 34.0 | 0.09% | 0.42 | 2,923,638,455 | |

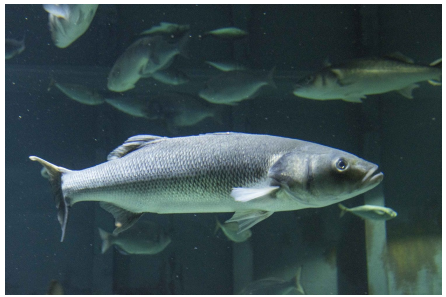
Hansen et al. 2026



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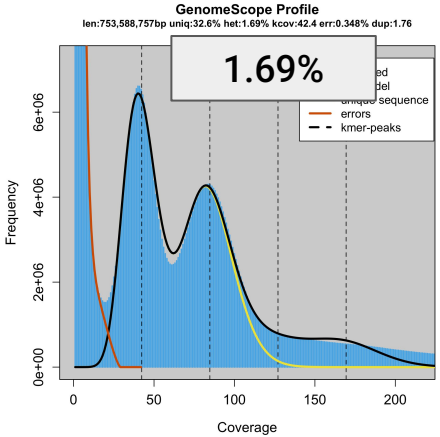
## Seabass



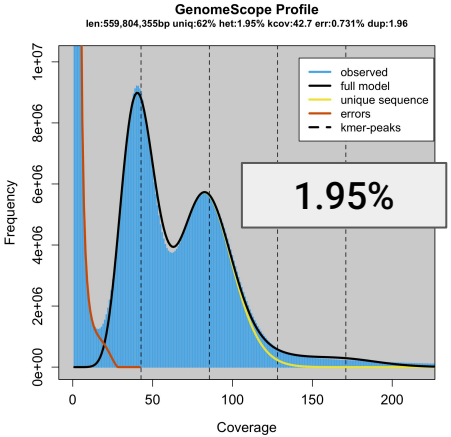





## Pear




## Oyster

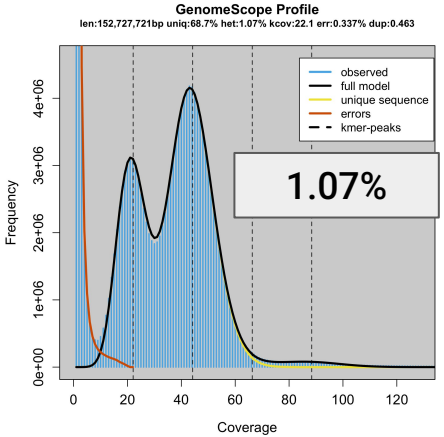




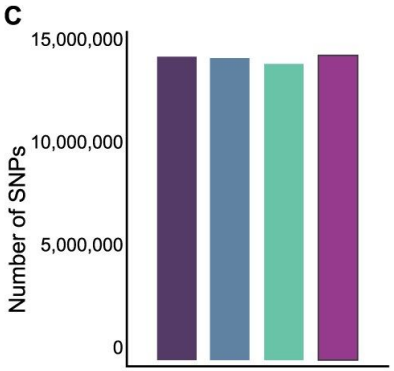


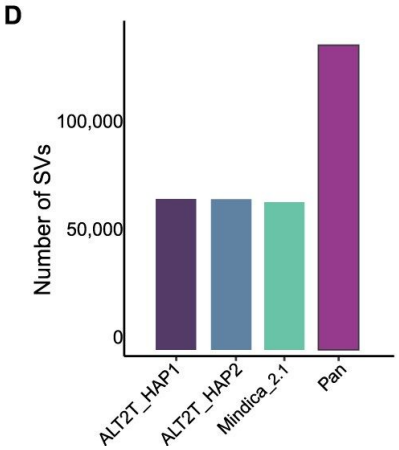


## Arabidopsis



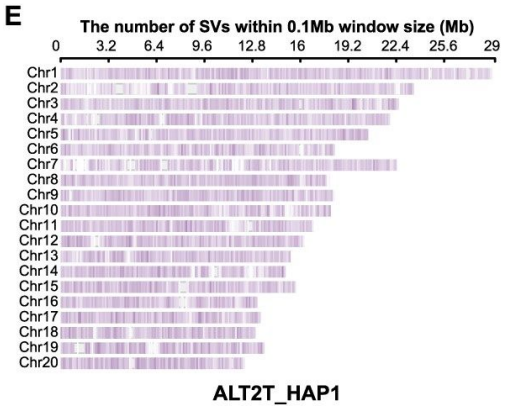
# Some variants are more likely to be missed with a single reference

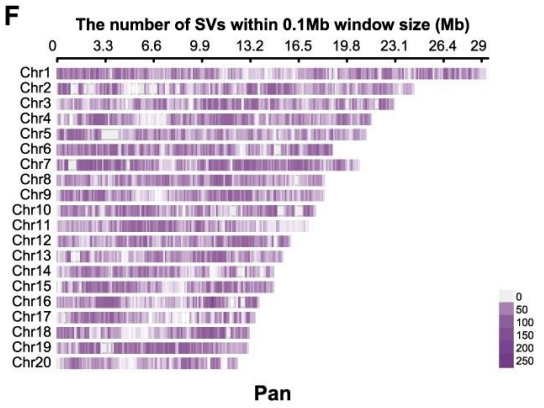





Authors found 50% more SVs with a mango pangenome than with any of the linear references. Even when using a T2T reference




Ahmed et al. 2025 https://academic.oup.com/hr/article/12/9/uhaf166/8179576

# Reference bias confounds analysis

- Hypervariable regions

JOURNAL ARTICLE

**Mapping Bias Overestimates Reference Allele Frequencies at the *HLA* Genes in the 1000 Genomes Project Phase I Data**

Débora Y C Brandt, Vitor R C Aguiar, Bárbara D Bitarello, Kelly Nunes, Jérôme Goudet, Diogo Meyer

*G3 Genes|Genomes|Genetics*, Volume 5, Issue 5, 1 May 2015, Pages 931–941,
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**Published:** 01 May 2015 **Article history**

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Version 2 Published: July 26, 2019 • https://doi.org/10.1371/journal.pgen.1008302

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- Allele specific effects

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Home > Genome Biology > Article

**Personalized and graph genomes reveal missing signal in epigenomic data**

Research | Open access | Published: 25 May 2020
Volume 21, article number 124 (2020) Cite this article

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- Hypervariable regions
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- Allele specific effects
- Epigenomic analysis/signaling
- Disease-associated variants

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New Results

**T2T-CHM13 reference genome reduces mapping bias and enhances alignment accuracy at disease-associated variants**

Ilaria Cherchi, Francesco Orlando, Orsetta Quaini, Marta Paoli, Yari Ciani, Francesca Demichelis

**doi:** https://doi.org/10.64898/2025.12.17.694618

This article is a preprint and has not been certified by peer review [what does this mean?].

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**d graph genomes reveal epigenomic data**

May 2020

Cite this article

# Quantifying reference bias is it's own challenge

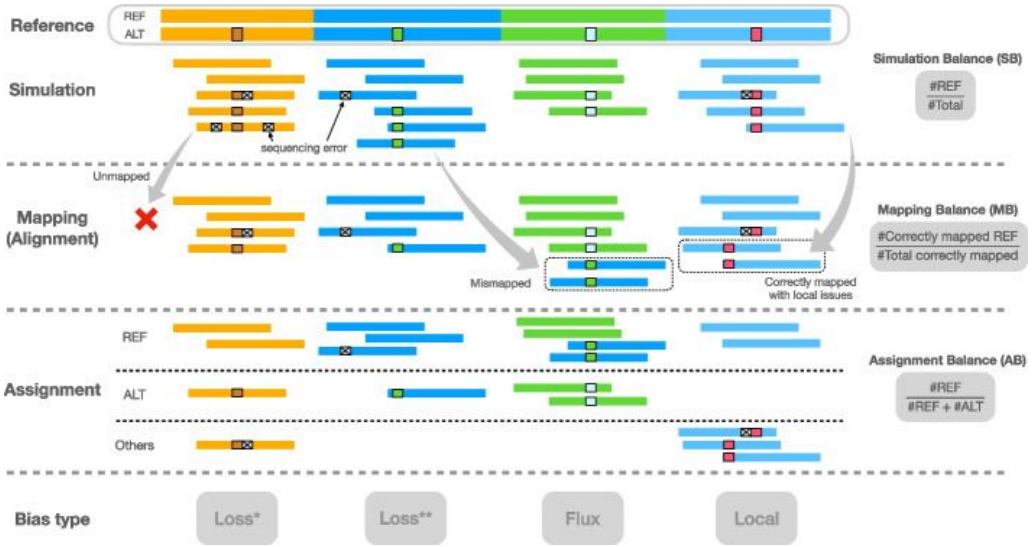
Genome Biol. 2024 Apr 19;25:101. doi: 10.1186/s13059-024-03240-8

## Measuring, visualizing, and diagnosing reference bias with biastools

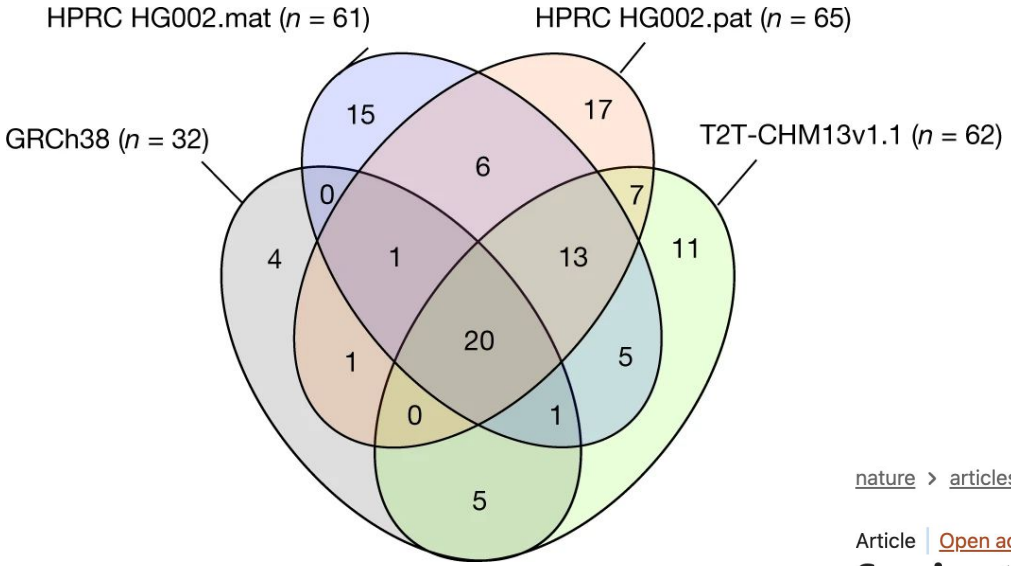
Mao-Jan Lin [1,✉], Sheila Iyer [1], Nae-Chyun Chen [1], Ben Langmead [1,✉]

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PMCID: PMC11027314 PMID: 38641647



# Even with multiple references, reference bias can still occur




**“Genes with no aligned transcript and thus presumed absent in the four main reference assemblies compared.”**

nature › articles › article

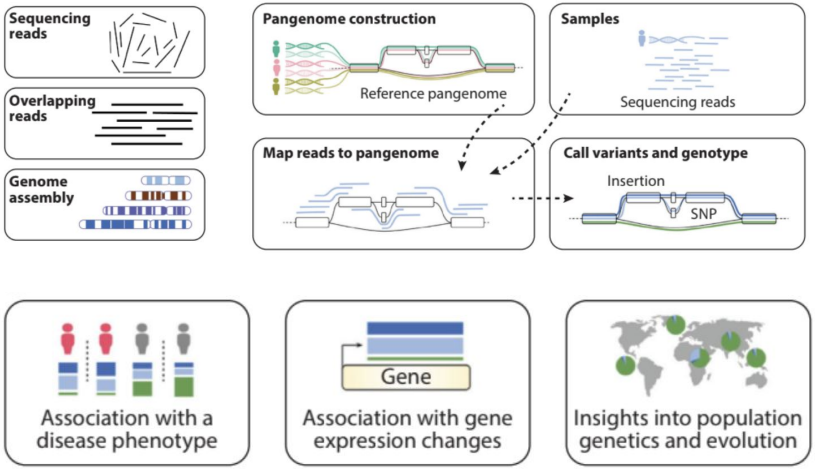
Article | Open access | Published: 19 October 2022

## Semi-automated assembly of high-quality diploid human reference genomes

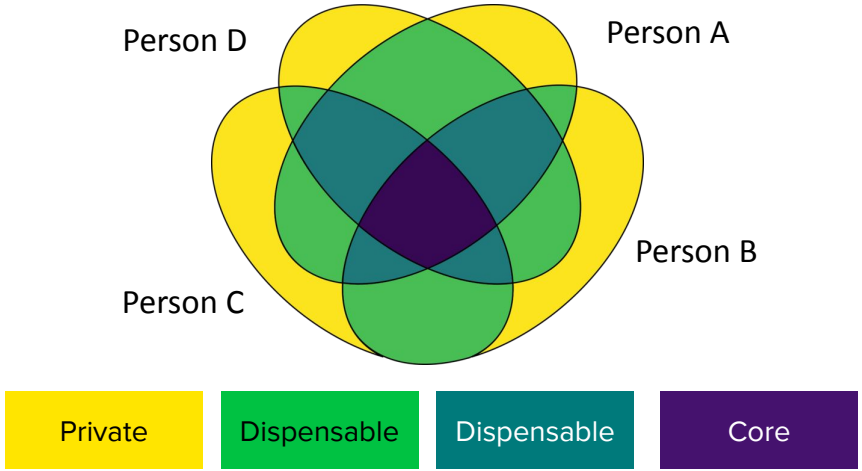
Erich D. Jarvis, Giulio Formenti, Arang Rhie, Andrea Guarracino, Chentao Yang, Jonathan Wood, Alan Tracey, Francoise Thibaud-Nissen, Mitchell R. Vollger, David Porubsky, Haoyu Cheng, Mobin Asri, Glennis A.

# Pangenomes mitigate reference bias

"A 'pangenome' is the collective whole-genome sequences of multiple individuals representing the genetic diversity of a species " - Human Pangenome Reference Consortium



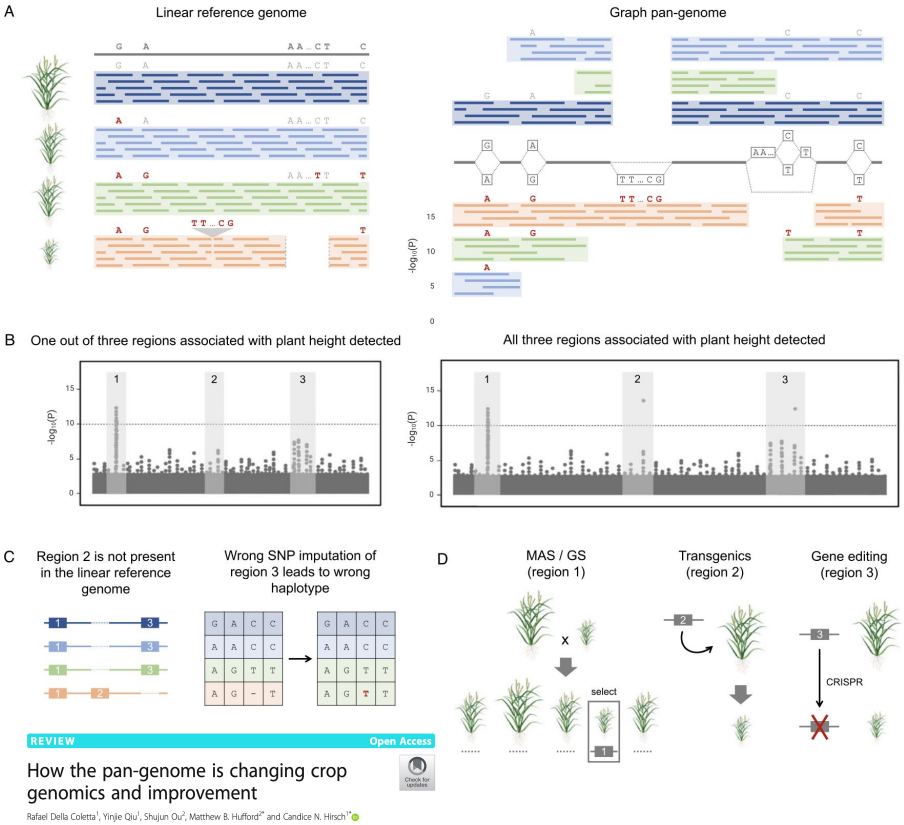

Taylor et al. AR Genomics and Human Genetics, 2024



# Pangenomes mitigate reference bias

Is one reference enough? No!

- One genome only represents one individual
- Hard to represent a whole species or clade with one individual
- Variation (especially large insertions) that aren't present in the reference will be hard to study
- One reference leads to **reference bias**, where the newly sequenced individual appears more similar to the reference than it actually is.



# We have a lot of data and need to organize it

| Dataset | Type | Reads | # accessions | Index/Seqs | Compress. |
|---|---|---|---|---|---|
| Sourmash Branchwater | Index | 1.4 Pbp | 682,688 | 7 TB | 191× |
| MetaGraph-SRA | Index | 3.3 Pbp | 1,891,328 | 7 TB | 473× |
| Pebblescout-SRA | Index | 3.7 Pbp | 4,141,058 | 171 TB | 21× |
| NCBI SRA (Dec 2023) | Raw data | 50.3 Pbp | 27,764,168 | 19 PB | 3× |
| Logan, Unitigs (v1) | Assembly | 48.3 Pbp | 27,311,279 | 2 PB | 24× |
| Logan, Contigs (v1.1) | Assembly | 44.1 Pbp | 26,788,829 | 315 TB | 140× |
| Logan-Search | Index | 43.9 Pbp | 23,404,655 | 1 PB | 44× |

Chikhi et al. “Logan: Planetary-Scale Genome Assembly Surveys Life’s Diversity”biorxiv 2025

# Pangenomes are useful because

1. A single reference does not capture sufficient genetic variation for many (most?) applications

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1. A single reference does not capture sufficient genetic variation for many (most?) applications

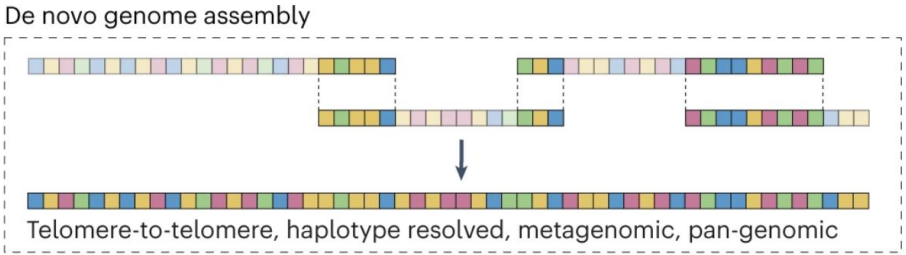
2. There is a LOT of data and pangenomes allow us to organize this data

# Outline

1. What is a pangenome?
2. Why do we need/want a pangenome?
3. Genome assemblies (the building blocks of pangenomes)
4. Representations of pangenomes
5. Examples of pan-genomes
6. Open questions in the field
7. What we will do with pangenomes in this course

# Genome assemblies are a hypothesis of the original nucleotide sequence, generated from shorter sequencing

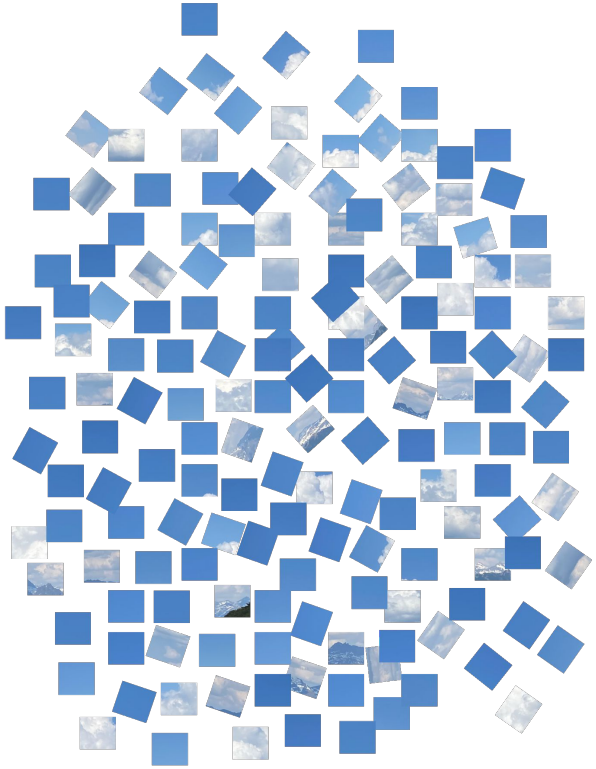


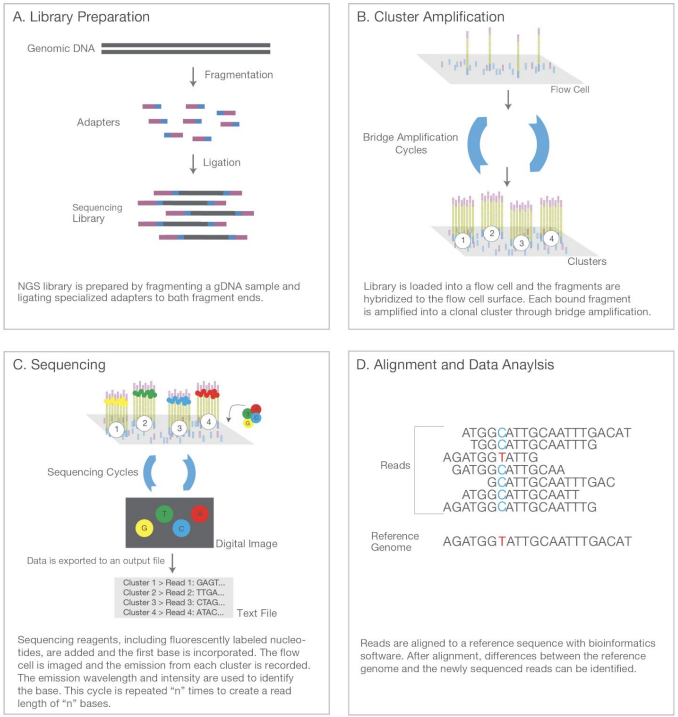

The first assembly of a human genome was a multi-billion dollar endeavour that took over 10 years to complete.

Kovaka S., Ou S., Jenike KM, Schatz MC. Nature Methods 2023

# Technological advances in sequencing have made complete genome assembly possible

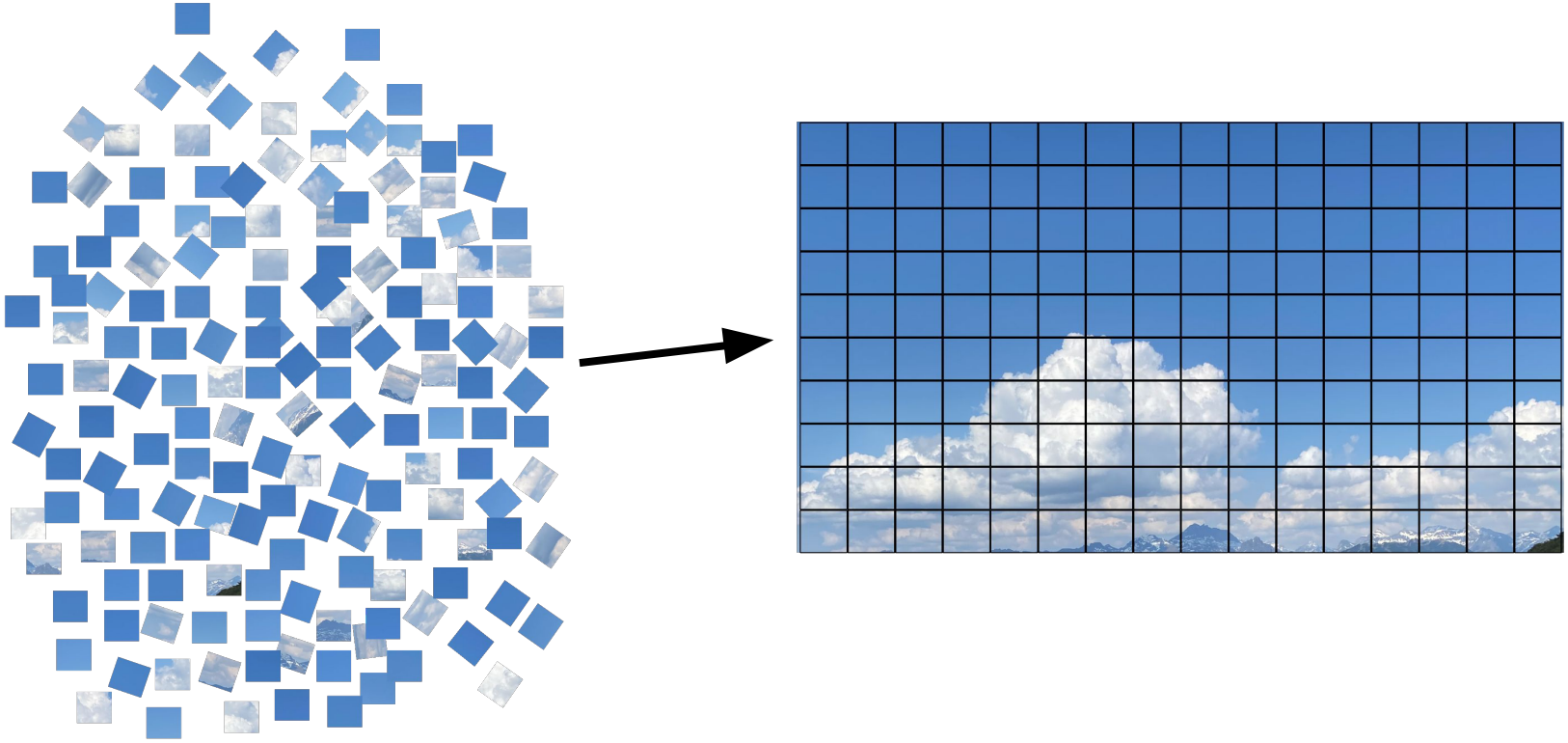


## Sequencing by synthesis



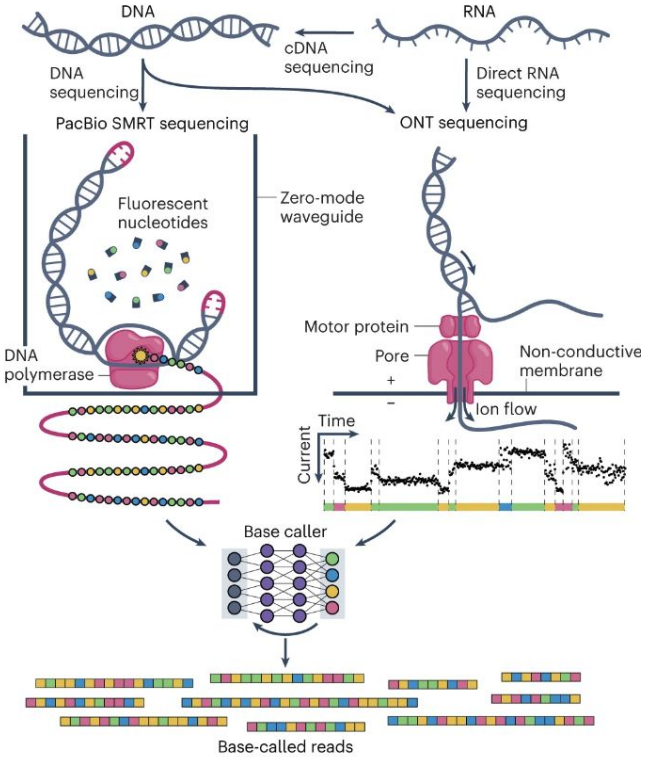

Illumina company website: https://www.illumina.com/science/technology/next-generation-sequencing.html

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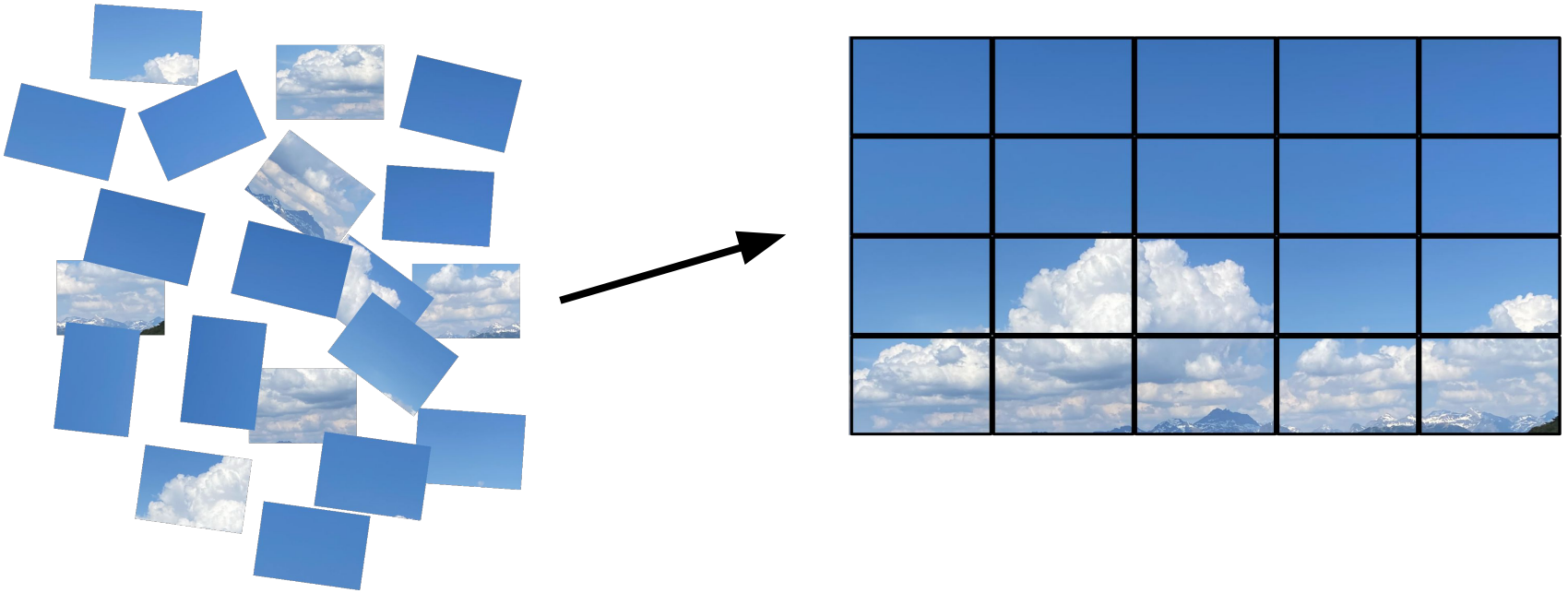
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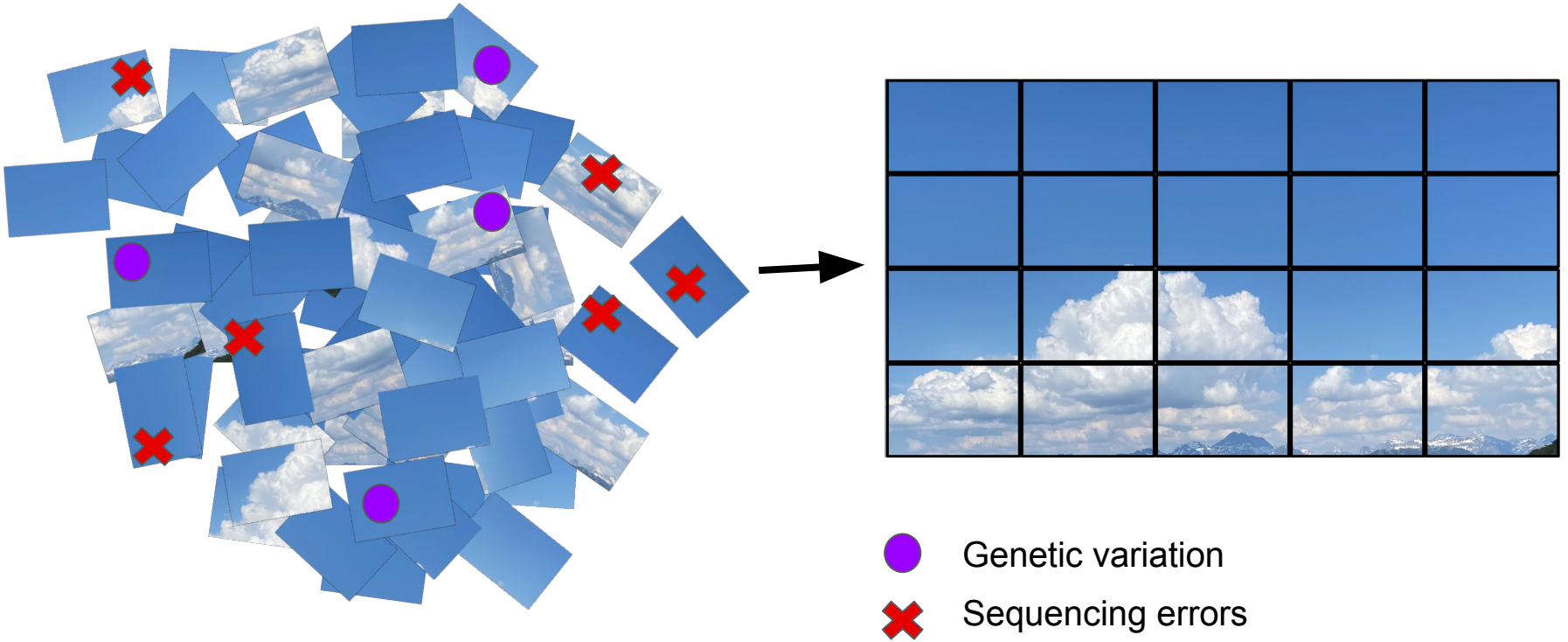



Kovaka et al. Nature Methods 2023

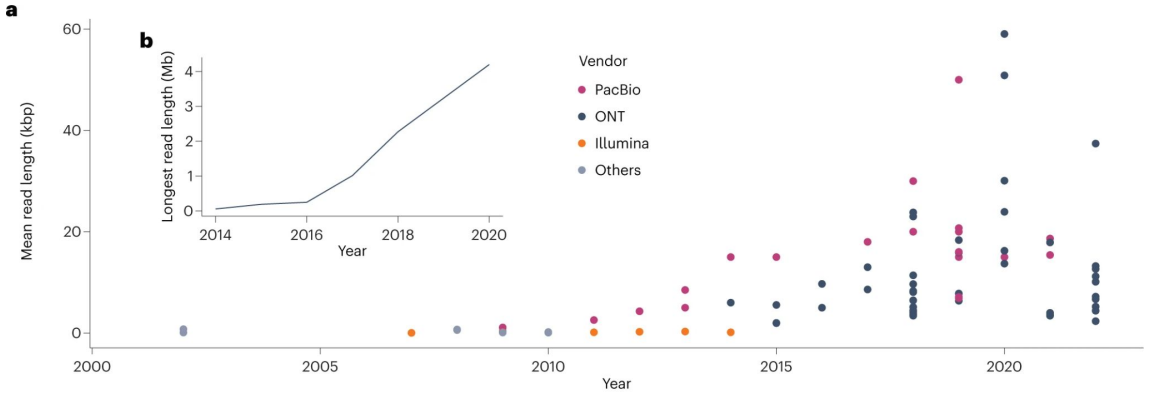
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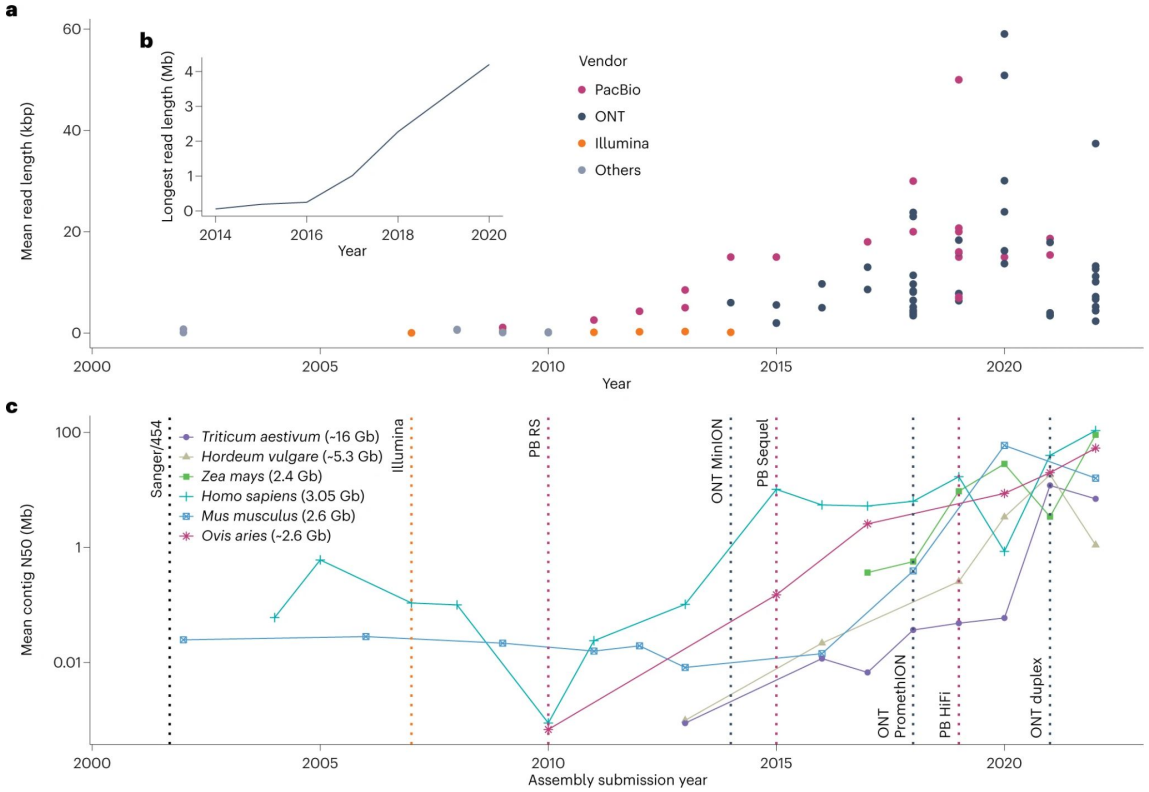


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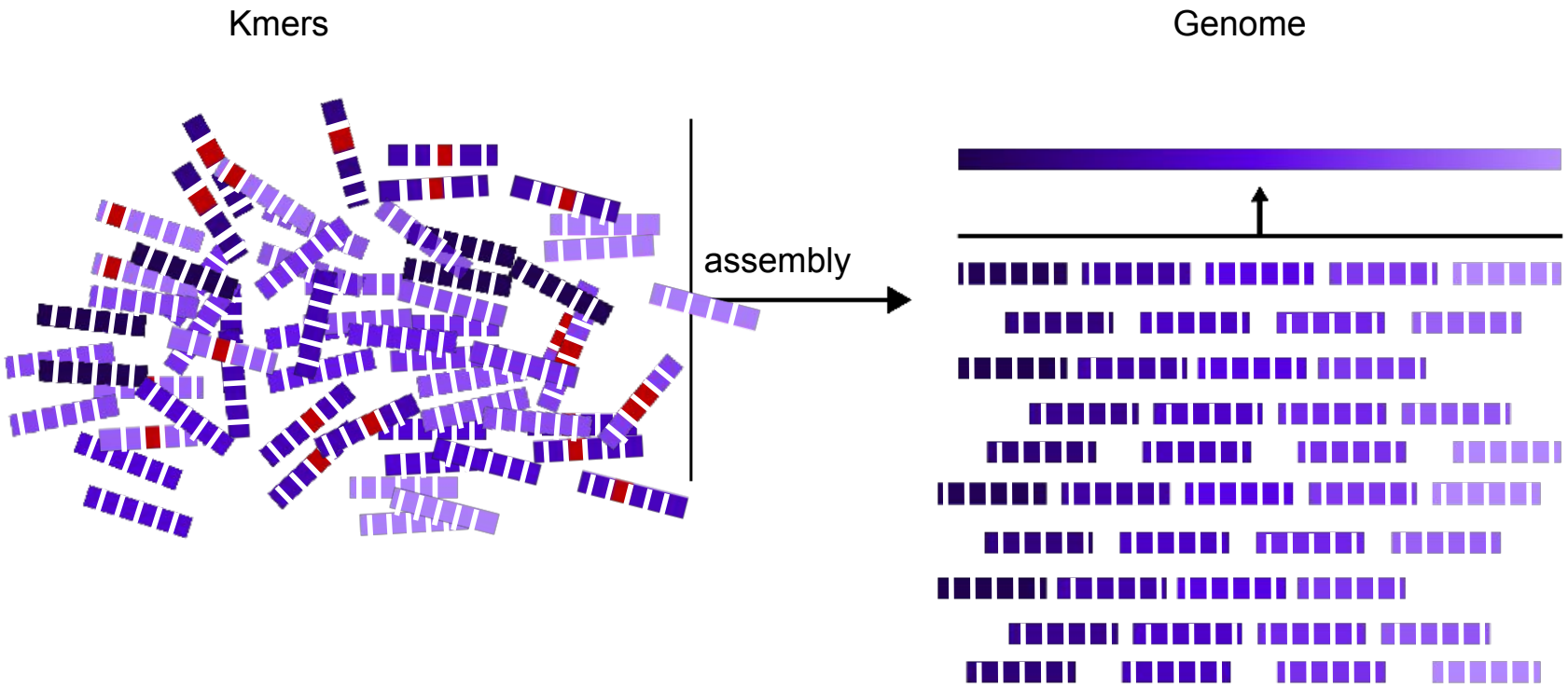

Kovaka et al. Nature Methods 2023

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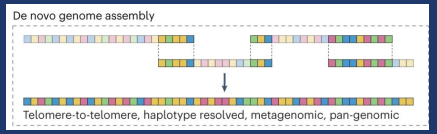



Kovaka et al. Nature Methods 2023

# Genome assembly example: Why breaking reads into smaller pieces is actually a good thing

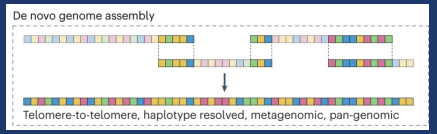


# Genome assemblers




**General advice: 30X coverage for diploids is recommended. ~15X per haplotype**

# Genome assemblers

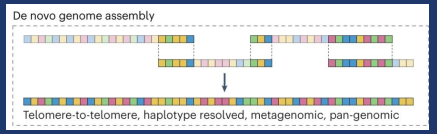



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Short reads/Illumina reads:

- De novo assembly is much more challenging with short reads
- Ideal for polishing genome assemblies and alignment based assemblies
- BWA-MEM, BowTie, and Minimap2

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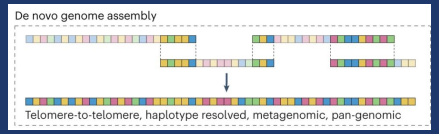
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Long reads

Oxford Nanopore (ONT):

- The longest reads (>100kbp)
- Flye, Hifiasm, Verkko, Shasta

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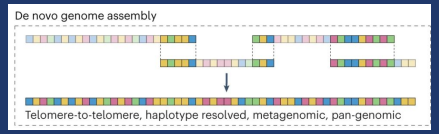
Oxford Nanopore (ONT):

- The longest reads (>100kbp)
- Flye, Hifiasm, Verkko, Shasta

PacBio:

- High fidelity reads, generally long (~10,000 bp)
- Hifiasm, Verrko, Canu

# Genome assemblers




Mini-Review

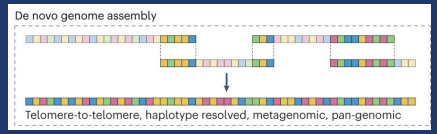
## A Hitchhiker's Guide to long-read genomic analysis

Medhat Mahmoud [1, 4], Daniel P. Agustinho [1, 4], Fritz J. Sedlazeck [1, 2, 3], Show affiliations

Published April 14, 2025. Vol 35 Issue 4, pp. 545-558. https://doi.org/10.1101/gr.279975.124

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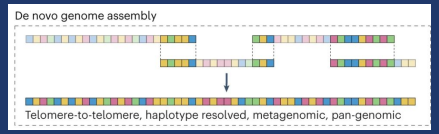
# Genome assemblers




| | | | | |
|---|---|---|---|---|
| Assembly (M) | Canu | ONT/PacBio | https://github.com/marbl/canu | Koren et al. 2017 |
| | Flye | ONT | https://github.com/mikolmogorov/Flye | Konings et al. 2024 |
| | hifiasm | ONT/PacBio | https://github.com/chhylp123/hifiasm | Cheng et al. 2021 |
| | Shasta | ONT | https://github.com/chanzuckerberg/shasta | Shafin et al. 2020 |
| | Verkko | ONT + PacBio + Illumina | https://github.com/marbl/verkko | Rautiainen et al. 2023 |

**Mahmoud et al. 2025**

# Genome assemblers




nature > articles > article

Article | Open access | Published: 27 April 2026

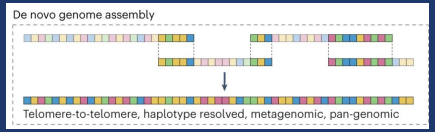
## Telomere-to-telomere assembly using HERRO-corrected Nanopore Simplex reads

Dominik Stanojević, Dehui Lin, Sergey Nurk, Paola Florez de Sessions & Mile Šikić

*Nature* **655**, 158–165 (2026) | Cite this article

Note for people with ONT only data. HERRO is a new tool to make really good assemblies from ONT only

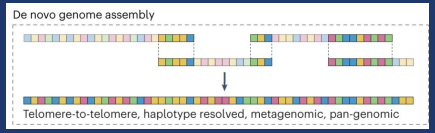
# Factors that impact genome assembly




> *The problem of sequence assembly can be compared to taking many copies of a book, passing each of them through a shredder with a different cutter, and piecing the text of the book back together just by looking at the shredded pieces. Besides the obvious difficulty of this task, there are some extra practical issues: the original may have many repeated paragraphs, and some shreds may be modified during shredding to have typos. Excerpts from another book may also be added in, and some shreds may be completely unrecognizable.*

https://en.wikipedia.org/wiki/Sequence_assembly

# Factors that impact genome assembly

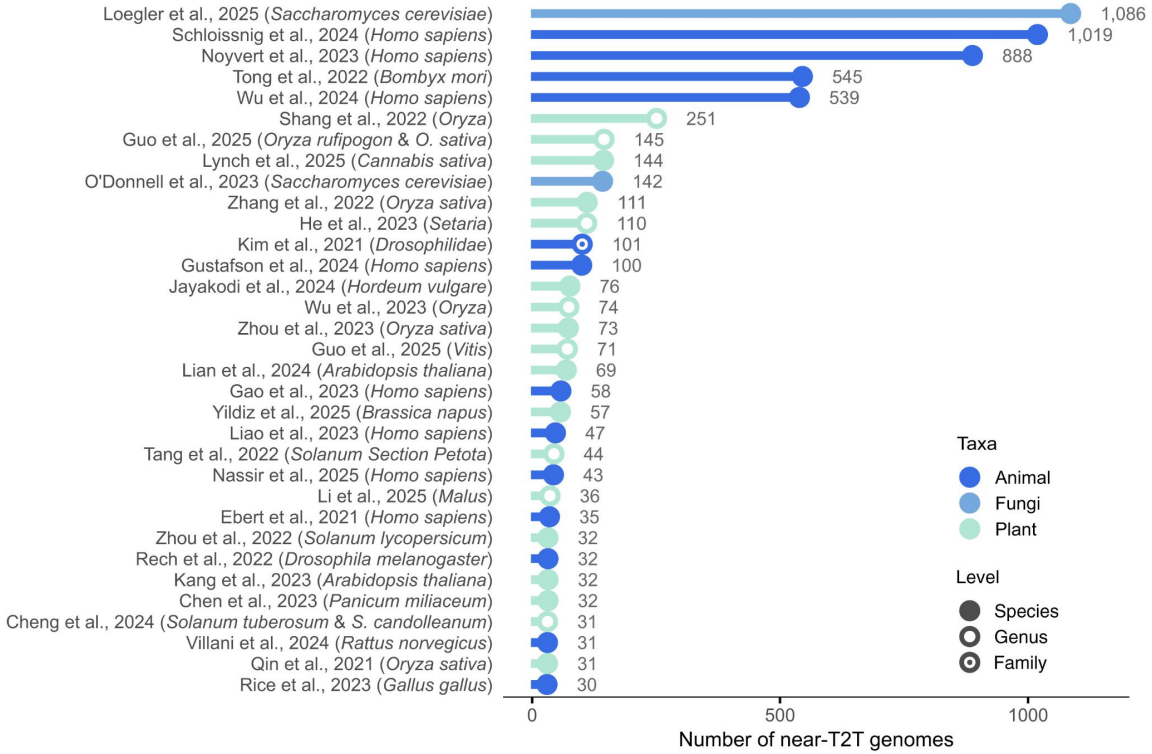



- Sequencing error
- Repetitive regions
- Polyploidy
- High rates of heterozygosity (sometimes)
- Read length

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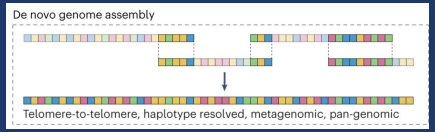
https://en.wikipedia.org/wiki/Sequence_assembly

# High quality genome assemblies are increasing!




Loegler et al. 2026 https://www.cell.com/cell-genomics/fulltext/S2666-979X(25)00323-4

# Genome assembly summary




- Long reads can enable near T2T and near complete genome assemblies
- Several genome assembly methods exist, the best option depends on the available data
  - Verkko
  - Hifiasm
  - Flye
- Most, but not all, T2T or near T2T assembly methods require multiple data types (HiFi, ONT, HiC/PoreC, ONT-UL)
- Genomes are the building blocks for pangenomes!

Great review by Heng Li and Richard Durbin on assembly methods:
https://www.nature.com/articles/s41576-024-00718-w

# Full (Telomere-to-telomere) genome assemblies are possible! Sometimes. With enough effort.

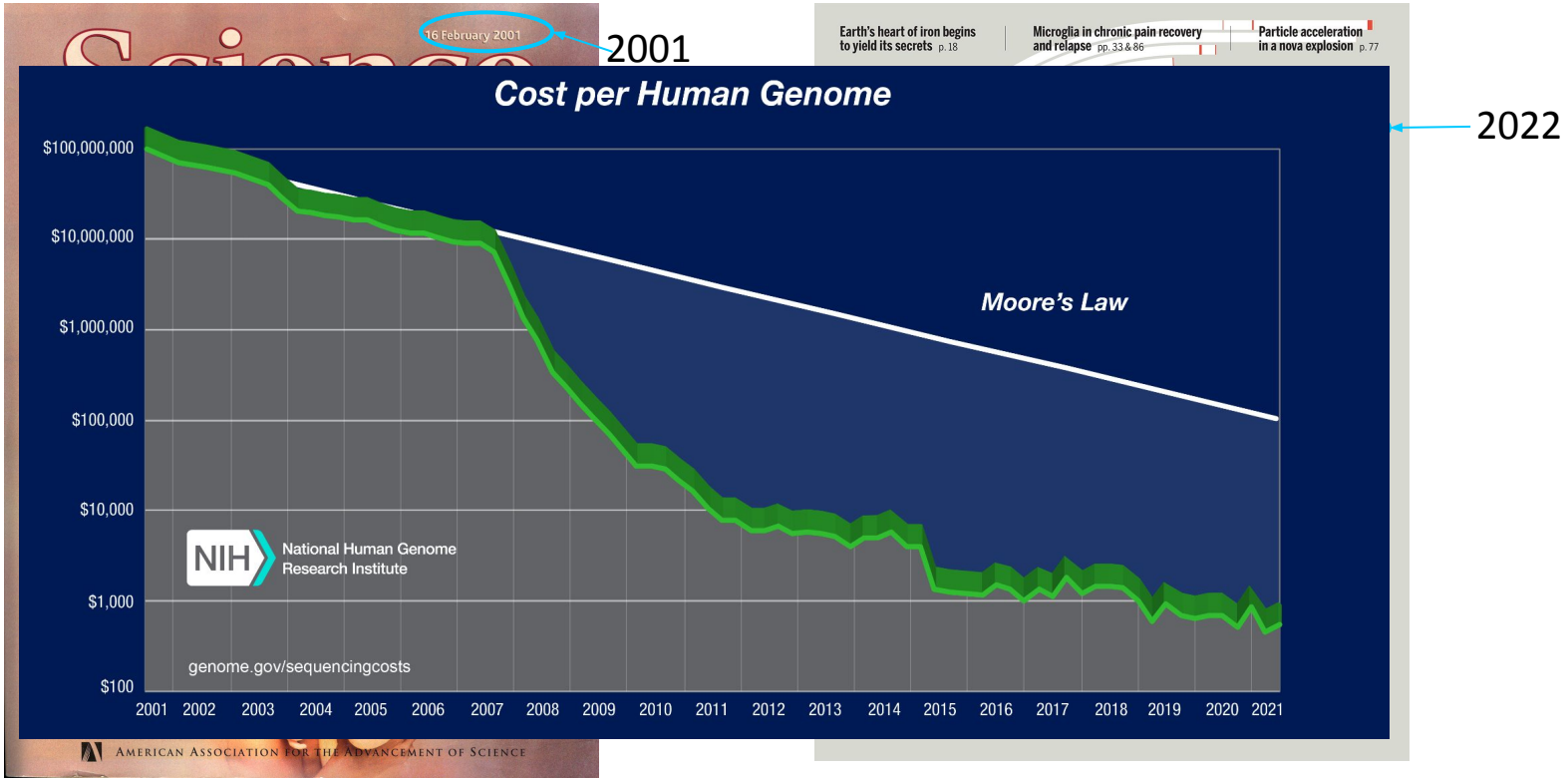



2001

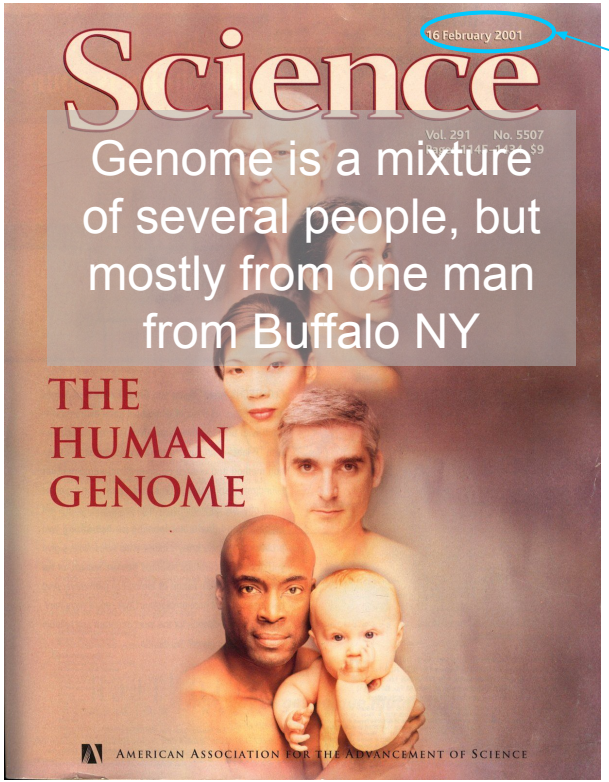



2022

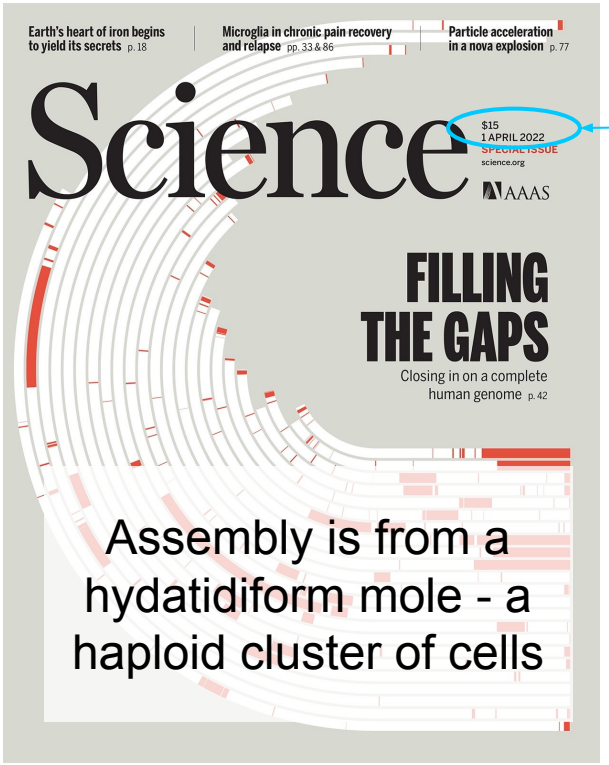
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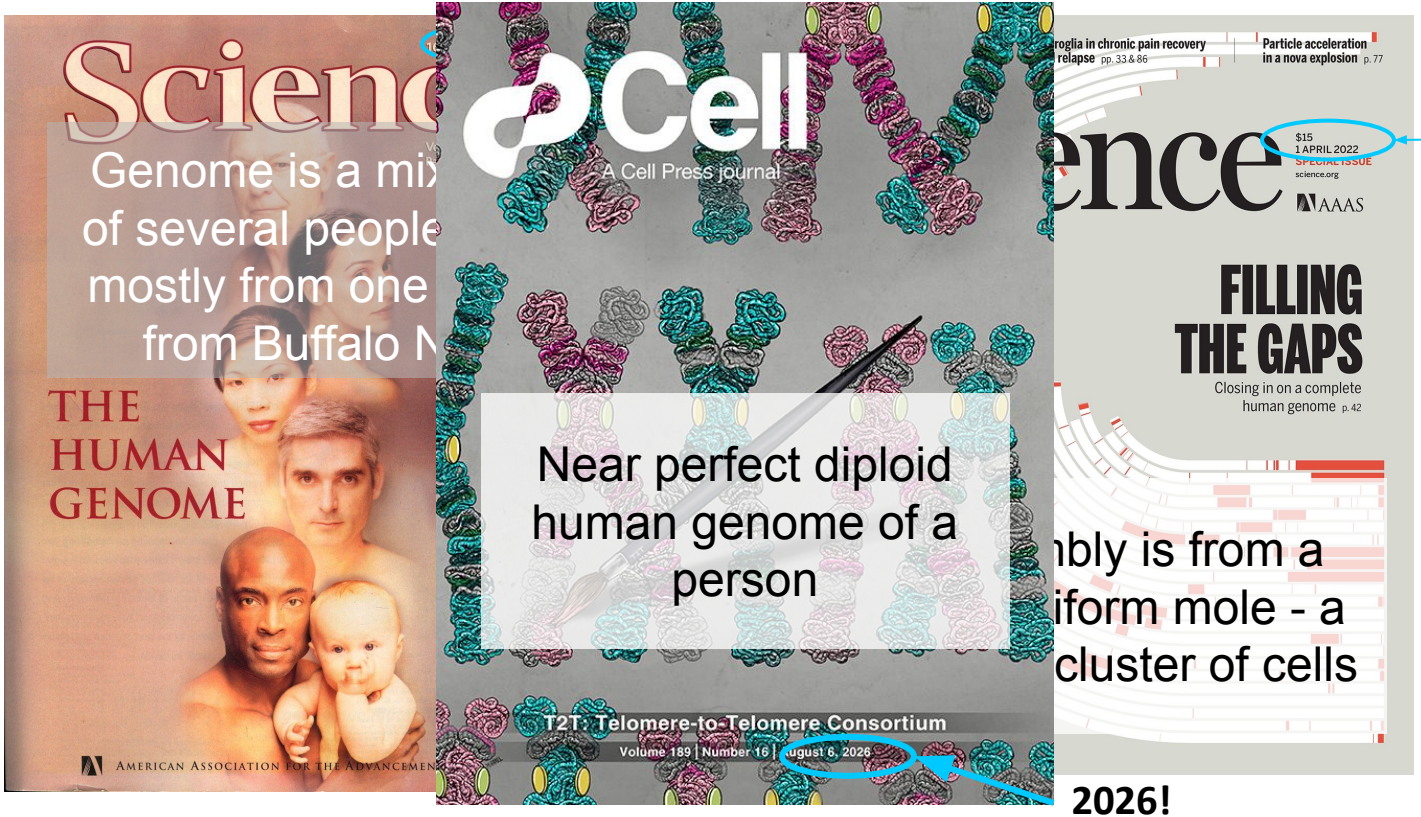



2001




2022

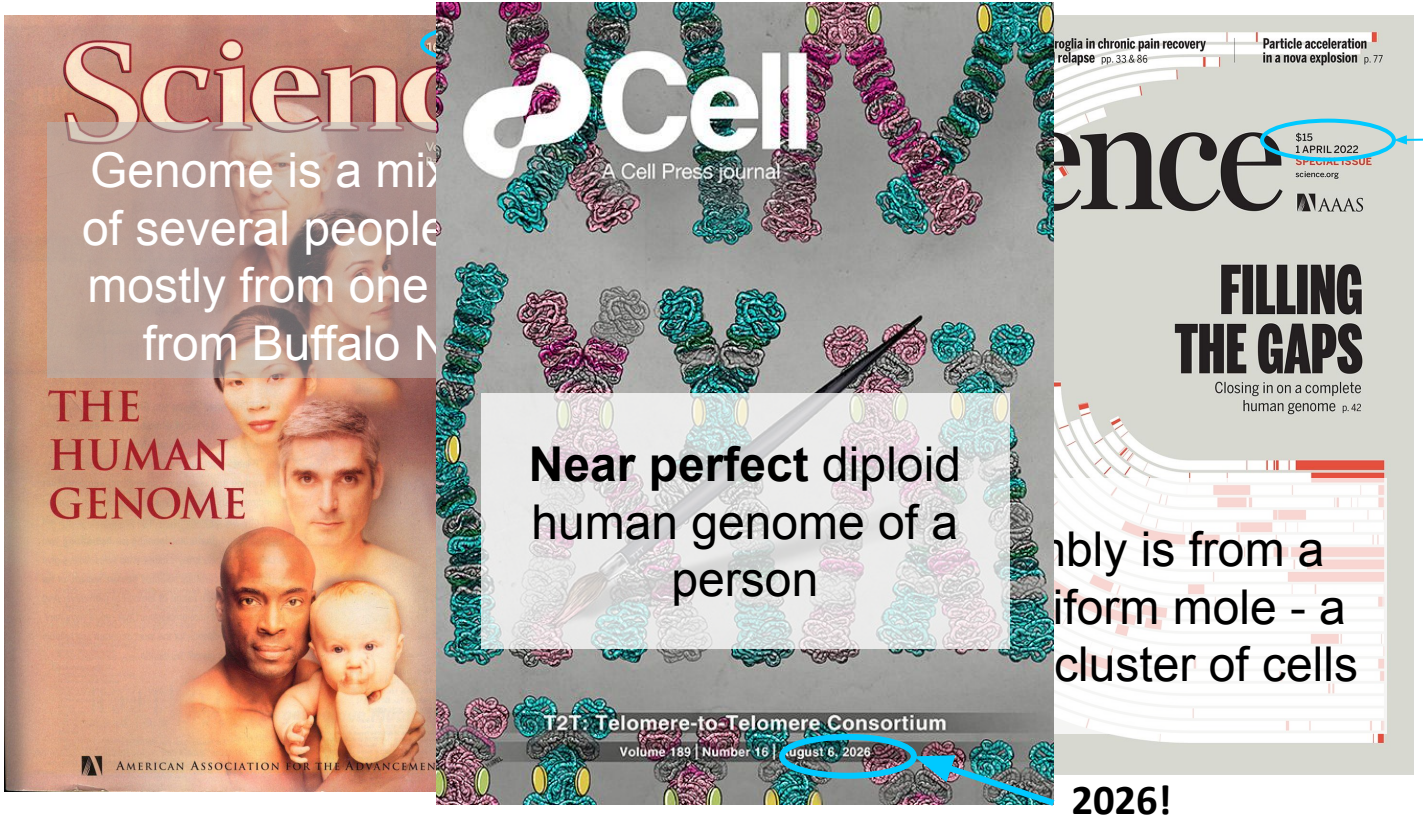
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2022

2026!

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2022

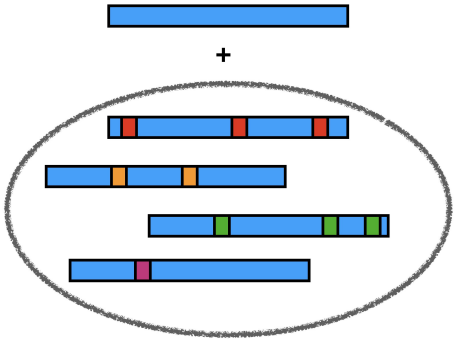
2026!

# Outline

1. What is a pangenome?
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# How to represent a collection of genomes?

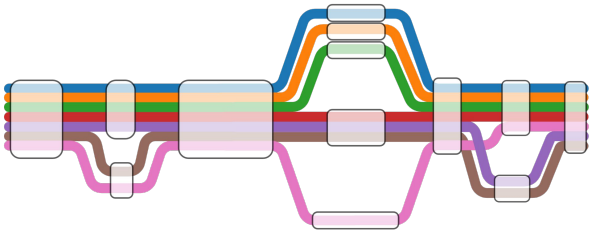
Depends on what purpose it serves!



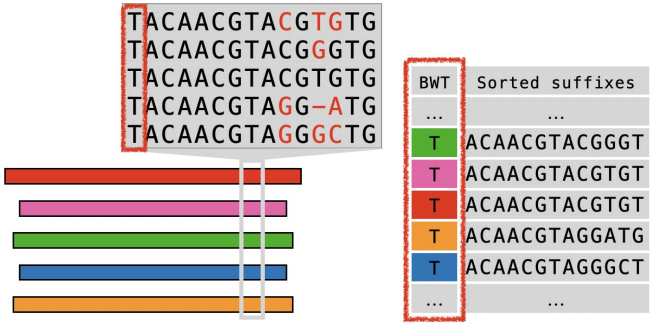

Reference genome
+ variants

| | G0 | G1 | G2 |
|---|---|---|---|
| act | 0 | 1 | 0 |
| cac | 1 | 1 | 0 |
| ccg | 0 | 0 | 1 |
| cga | 0 | 0 | 1 |
| gac | 0 | 0 | 0 |
| gga | 0 | 0 | 1 |
| tca | 1 | 0 | 0 |

k-mer index



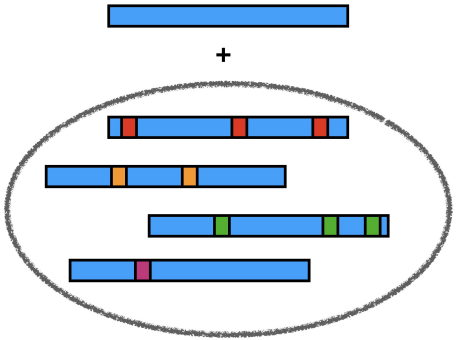
sequence graphs




Full-text index

# Compression

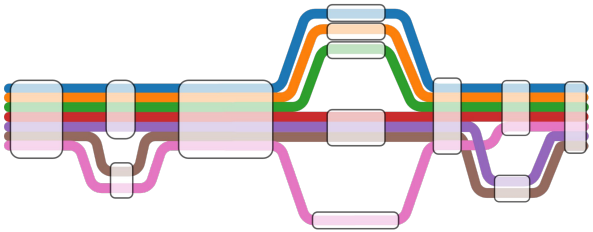
What do these all have in common? Compression!



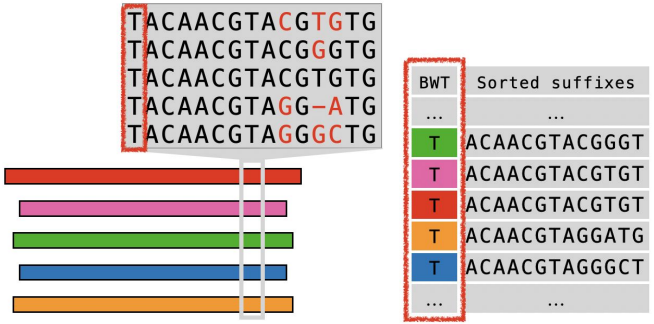
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| gac | 0 | 0 | 0 |
| gga | 0 | 0 | 1 |
| tca | 1 | 0 | 0 |

k-mer index



sequence graphs



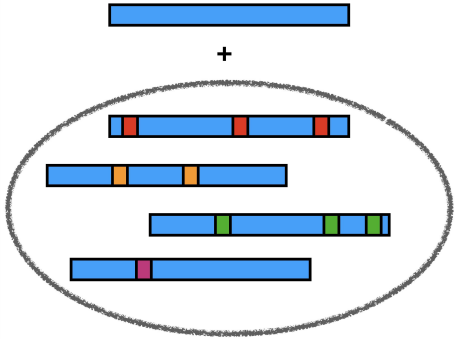

Full-text index

# Catalog of variation across a pangenome

Often represented as a collection of vcfs (variant calling files), which contain “diffs”

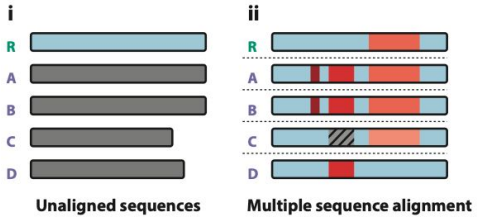
Generally is *not* lossless (we can’t represent complex variants)

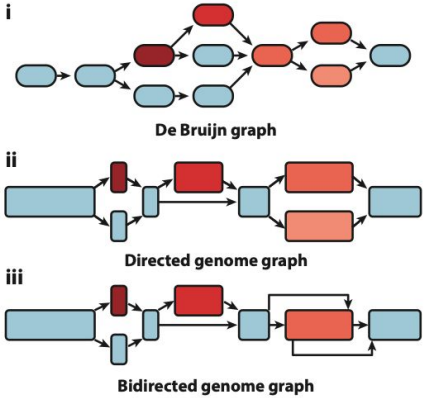
Involves *aligning* genomes, either all together or individually against the ref




**Reference genome + variants**

# Graph based pangenomes




*Annual Review of Genomics and Human Genetics*

Pangenome Graphs

Jordan M. Eizenga,[1] Adam M. Novak,[1]
Jonas A. Sibbesen,[1] Simon Heumos,[2] Ali Ghaffaari,[3,4,5]
Glenn Hickey,[1] Xian Chang,[1] Josiah D. Seaman,[6,7]
Robin Rounthwaite,[1] Jana Ebler,[3,4,5]
Mikko Rautiainen,[3,4,5] Shilpa Garg,[8,9] Benedict Paten,[1]
Tobias Marschall,[3,4] Jouni Sirén,[1] and Erik Garrison[1]

- Variation graph
  sequences = nodes, path = genome

- Requires an underlying multiple sequence alignment

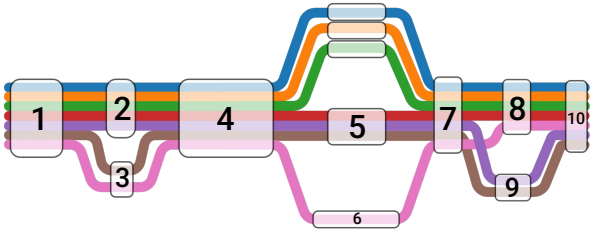
- Bubbles (“snarls”) represent sites of variation

# Graph based pangenomes



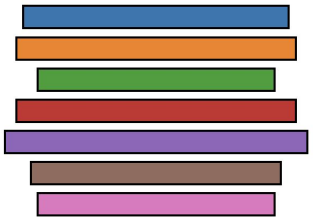
The “pangenome” is the set of walks through the graph

The graph is just a representation of an MSA!




pangenome graph

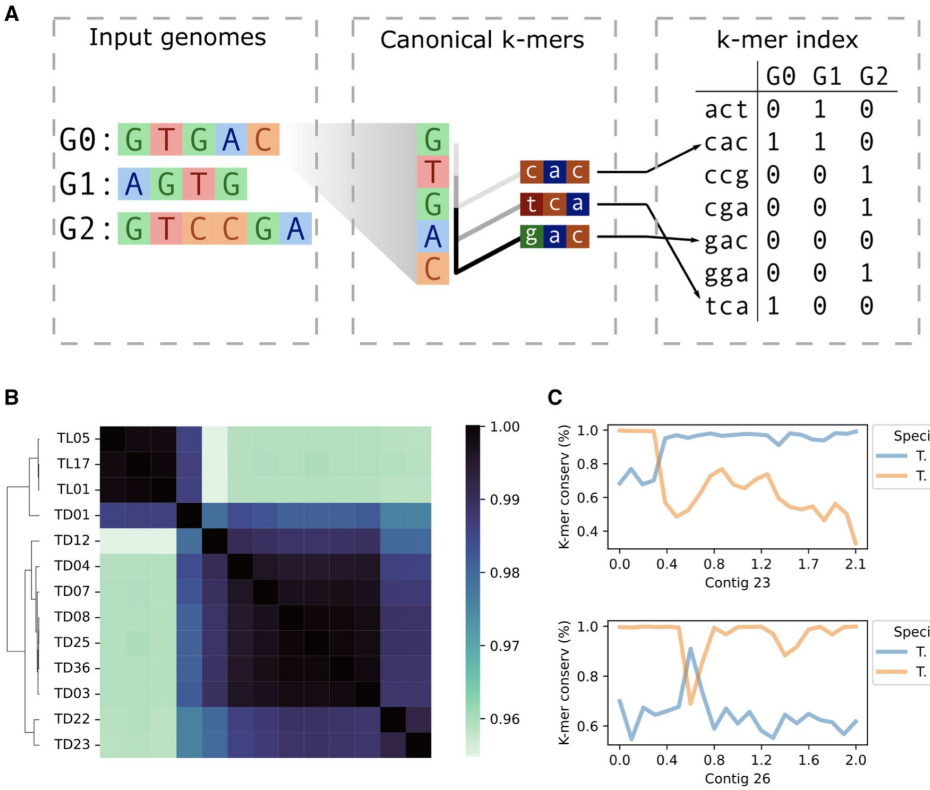
vs.



pangenome

1 -> 3 -> 4 -> 6 -> 7 -> 8 -> 10

# k-mer index



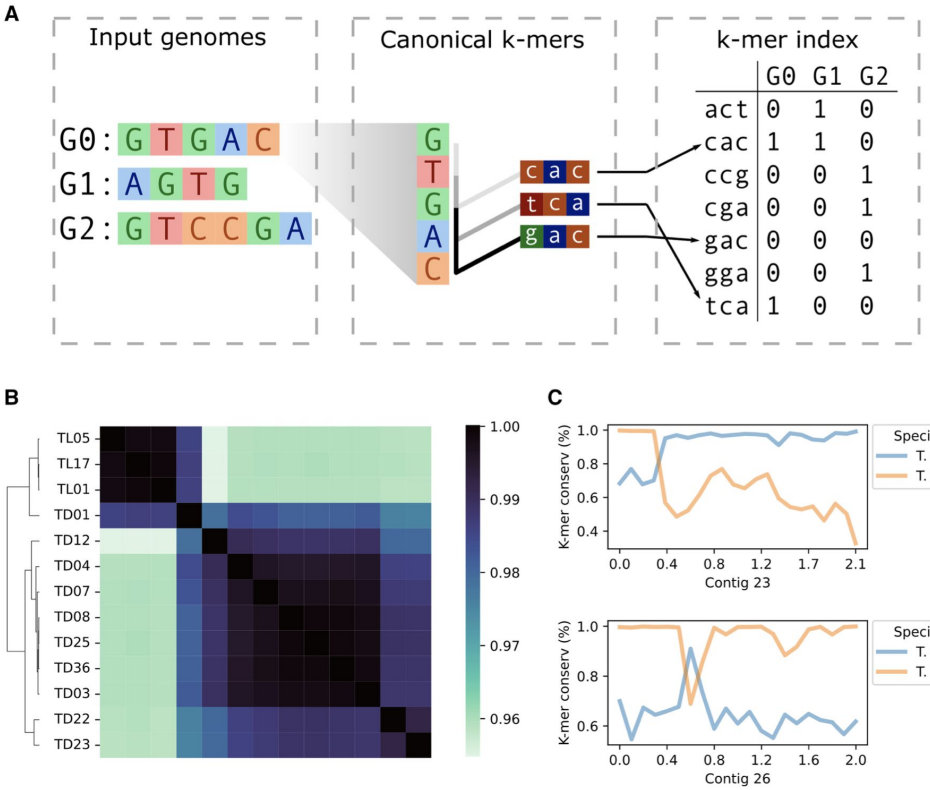

JOURNAL ARTICLE

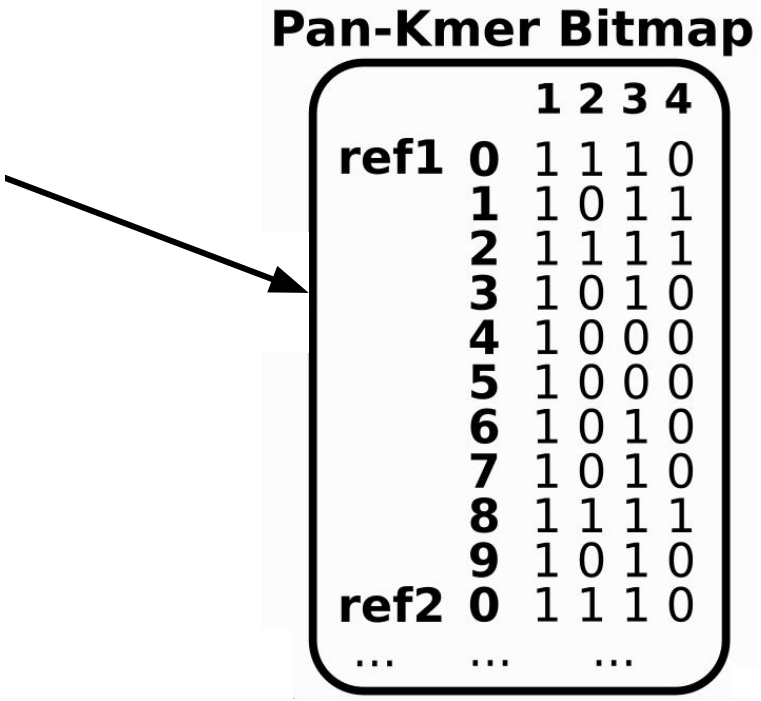
## PanKmer: *k*-mer-based and reference-free pangenome analysis

Anthony J Aylward, Semar Petrus, Allen Mamerto, Nolan T Hartwick, Todd P Michael

*Bioinformatics*, Volume 39, Issue 10, October 2023, btad621,
https://doi.org/10.1093/bioinformatics/btad621
**Published:** 16 October 2023 **Article history** ▾

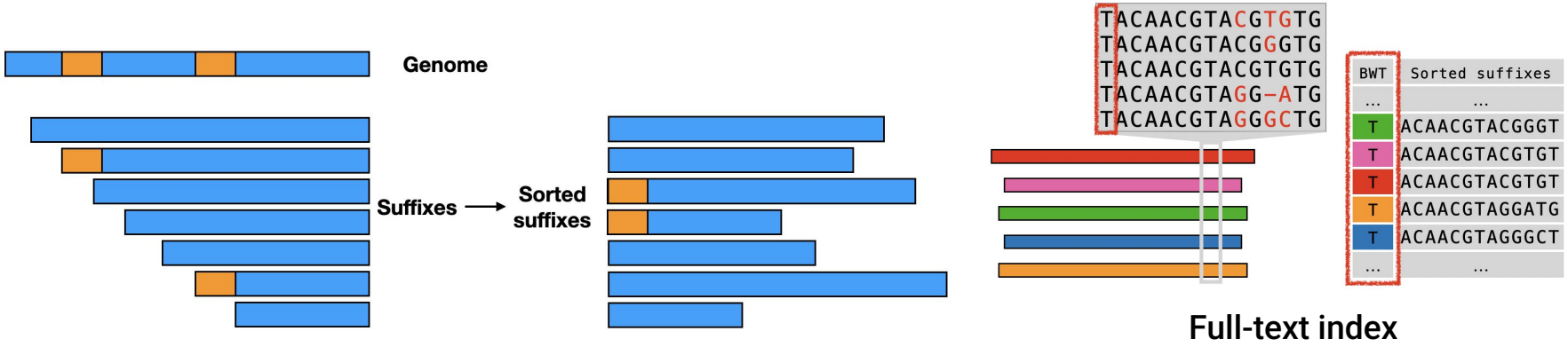
# Pan-k-mer analysis



# Full text indexes

- A query-able index of the genomes for searching
- Core to tools like **Bowtie2, BWA-MEM,** and **Mummer4**
- Frequently based on the Burrows-Wheeler Transform (BWT) or suffix array/tree

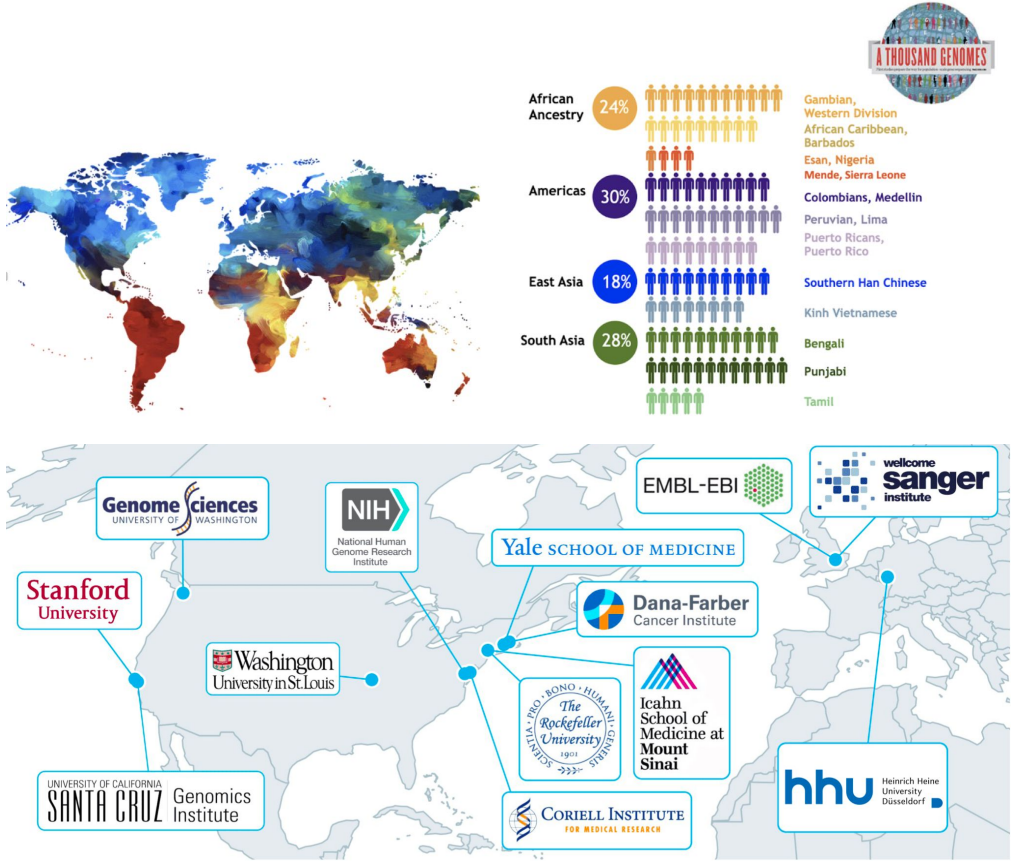


# Outline

1. What is a pangenome?
2. Why do we need/want a pangenome?
3. Genome assemblies (the building blocks of pangenomes)
4. Representations of pangenomes
5. Examples of pan-genomes
6. Open questions in the field
7. What we will do with pangenomes in this course

# Examples of pangenomes

**Human Pangenome Reference Consortium**



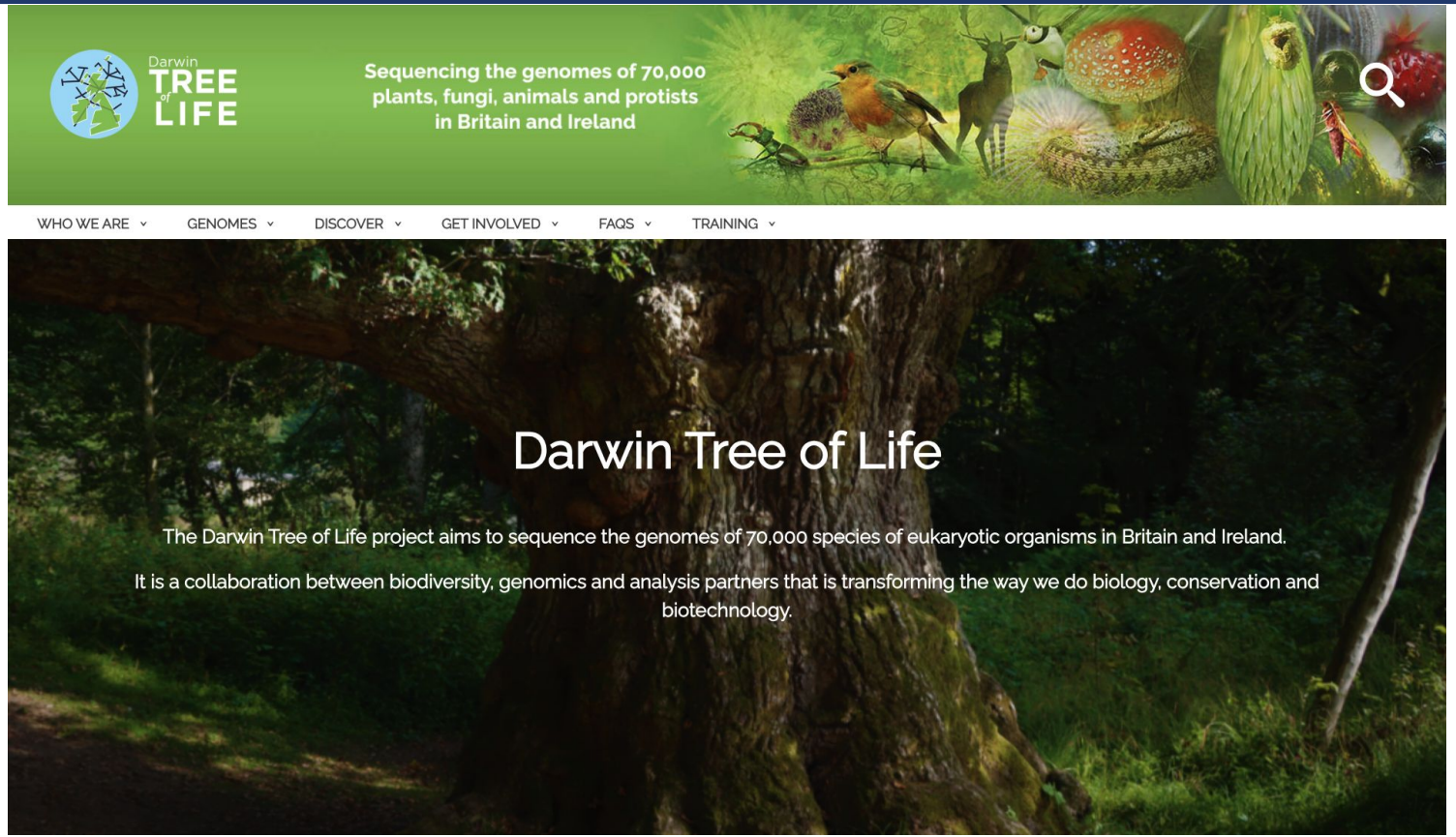
# Examples of pangenomes

VGP VERTEBRATE GENOMES PROJECT

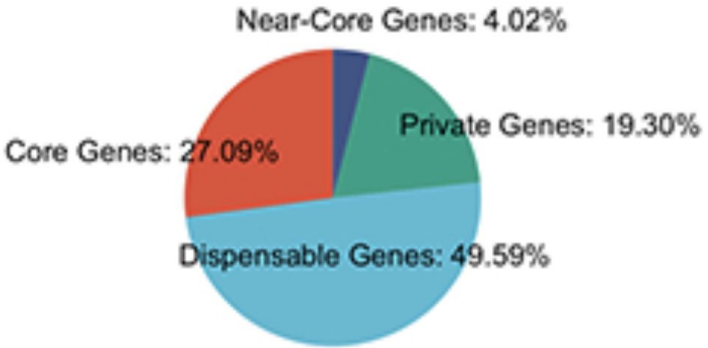
A PROJECT OF THE G10K CONSORTIUM



# Examples of pangenomes



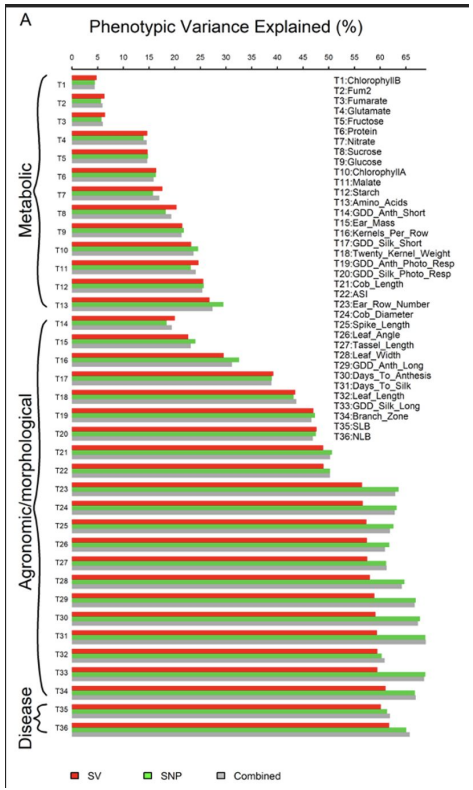
# Plant pangenome examples: Maize




Science. Author manuscript; available in PMC 2022 Feb 6.
PMCID: PMC8733867
*Published in final edited form as:*
NIHMSID: NIHMS1753799
Science. 2021 Aug 6; 373(6555): 655–662.
PMID: 34353948
doi: 10.1126/science.abg5289

*De novo* assembly, annotation, and comparative analysis of 26 diverse maize genomes

Matthew B. Hufford,[1] Arun S. Seetharam,[1,2] Margaret R. Woodhouse,[3] Kapeel M. Chougule,[4] Shujun Ou,[1] Jianing Liu,[5] William A. Ricci,[6] Tingting Guo,[8] Andrew Olson,[4] Yinjie Qiu,[9]




- 26 maize genomes picked because they are founders.
- 27% of the genes are core, found in all species.
- 19% of the genes are unique to an individual.
- Both SVs (>100bp, by their definition) and SNPs (<100bp) impact phenotype.
- SVs had a greater impact on disease associated phenotypes compared to metabolic phenotypes.

# Tomato pangenome with 100 tomatoes

Cell

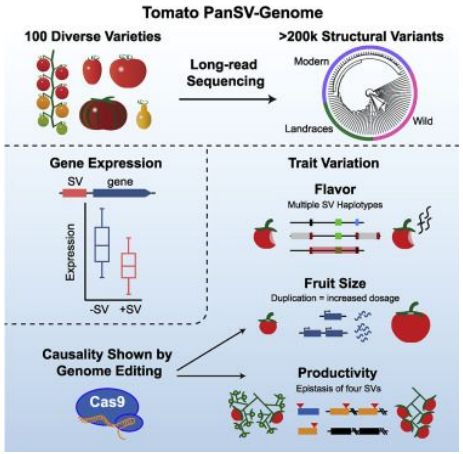
CellPress

Volume 182, Issue 1, 9 July 2020, Pages 145-161.e23

Article

## Major Impacts of Widespread Structural Variation on Gene Expression and Crop Improvement in Tomato

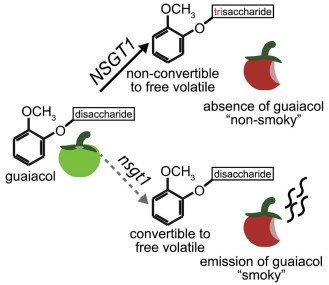
Michael Alonge [1 21], Xingang Wang [2 21], Matthias Benoit [2 3], Sebastian Soyk [2 19], Lara Pereira [4], Lei Zhang [4], Hamsini Suresh [2], Srividya Ramakrishnan [1], Florian Maumus [5], Danielle Ciren [2], Yuval Levy [6], Tom Hai Harel [6], Gili Shalev-Schlosser [6], Ziva Amsellem [6], Hamid Razifard [7 8], Ana L. Caicedo [7 8], Denise M. Tieman [9], Harry Klee [9], Melanie Kirsche [1], Sergey Aganezov [1] ... Zachary B. Lippman [2 3 22]



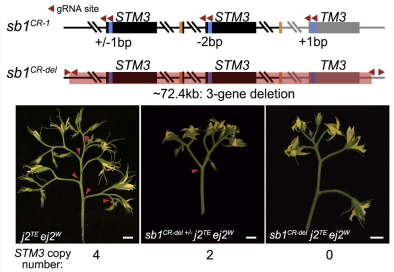



# Structural variants identified in the tomato pangenome influence gene expression and multiple important phenotypes

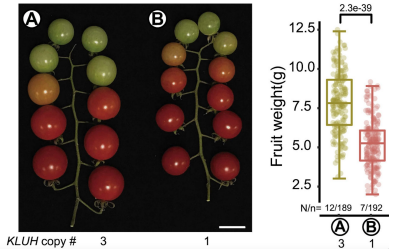
## Natural variation in smoky flavor




## Plant productivity




## Fruit weight






## Global crop production

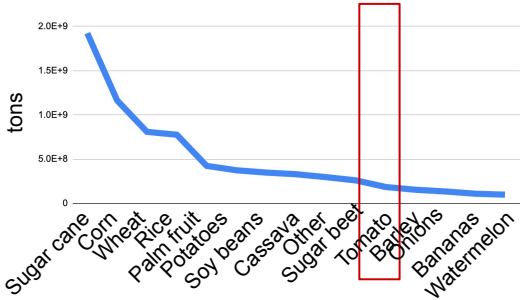





Photo taken by Mackenzie Kovaka

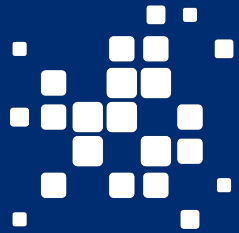
# Some open questions in the field

- How do you store pan-genomes efficiently?
- How do you visualize a pan-genome?
- Standardized pan-genome format
- How to move beyond a gene-centric analysis

# What we will do with pangenomes in this course

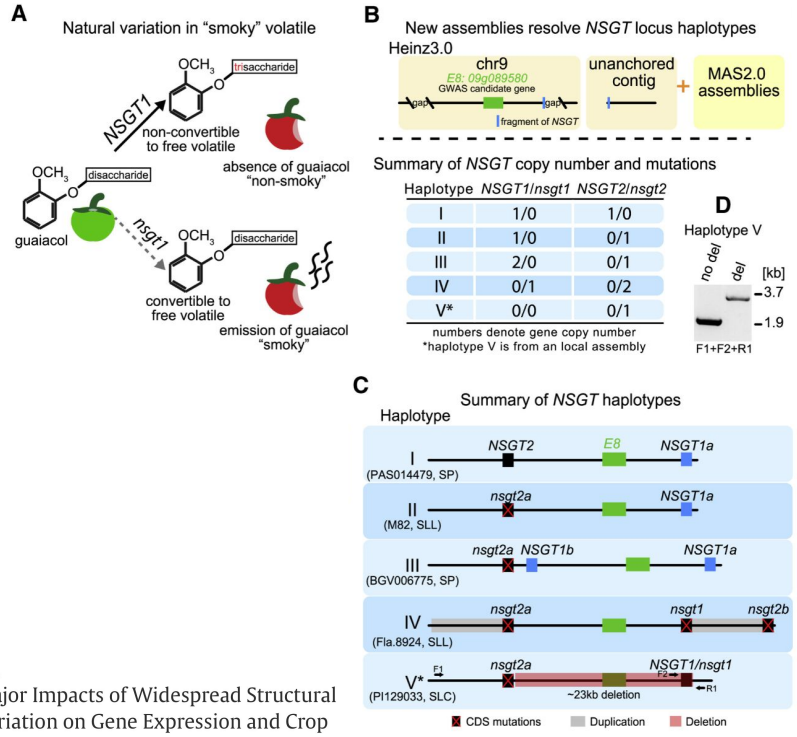
- Build a pangenome multiple ways and compare the output
- Compare pangenomes from different origins (human, plants, yeast)
- Assess runtime and memory requirements for different pangenome approaches
- Review recent pangenome applications
- Apply this to a problem of your choosing on Friday

# Questions?



wellcome sanger institute

# Plant pan-genome examples: Tomato




Article

Major Impacts of Widespread Structural Variation on Gene Expression and Crop Improvement in Tomato

Michael Alonge [1 21], Xingang Wang [2 21], Matthias Benoit [2 3], Sebastian Soyk [2 19], Lara Pereira [4], Lei Zhang [4], Hamsini Suresh [2], Srividya Ramakrishnan [1], Florian Maumus [5], Danielle Ciren [2],

- Smoky aromatic trait pathway: occurs when a NSGT1 does NOT
  - NSGT1 converts smoky volatiles to non-cleavable triglycerides
  - Mutations in NSGT1 (nsgt1) mean the smoky volatiles are emitted
- Found five different haplotypes
  - Some had coding variants
  - One had duplication
  - One had a deletion
- All impacted the smoky phenotype